



# Chapter 3

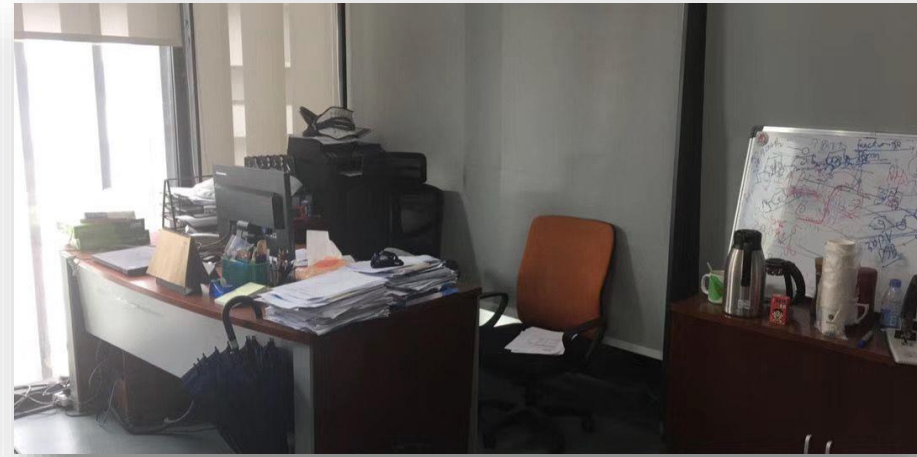
## The Second Law

**The Second Law** may be expressed in terms of the **entropy**, a state function that lets us assess whether one state is accessible from another by a **spontaneous change**. the Second Law uses the **entropy to identify** the **spontaneous changes** among those permissible changes.

**The First Law** led to the introduction of the **internal energy**, a state function that lets us assess whether a change is **permissible**; only those change may occur for which the  $U$  of an isolated system remains constant. First Law uses the **internal energy to identify** permissible changes;



# 1. What is the spontaneous process?



Spontaneous change as time passes

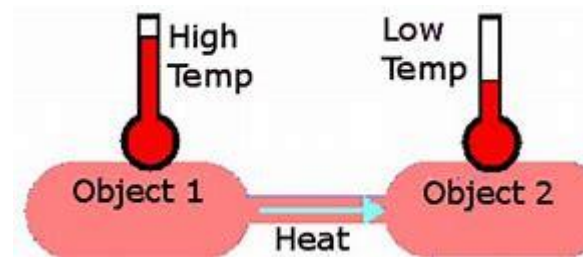


Requires energy input

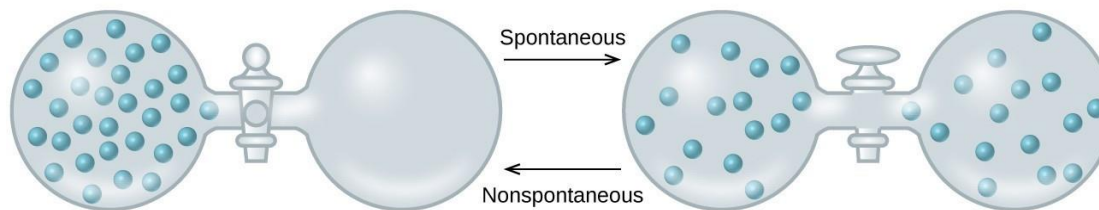


Nonspontaneous

**Spontaneous processes have a natural tendency to occur in one direction under a given set of conditions.**

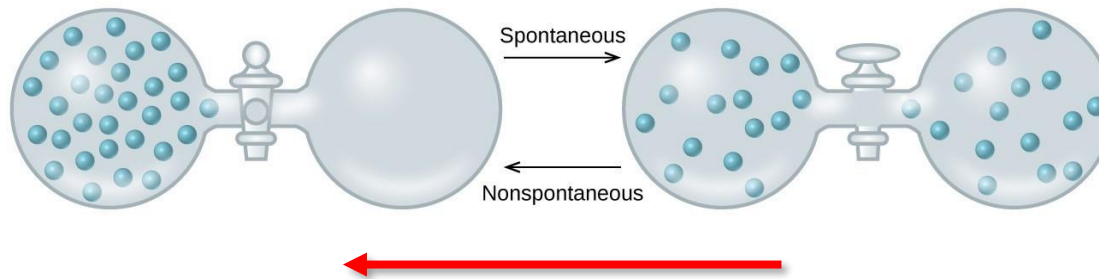


**Heat: from high temperature to low temperature**



**Gas: from high pressure to low pressure**

**A nonspontaneous process, on the other hand, will not take place unless it is “driven” by the continual input of energy from an external source.**



**Gas: from low pressure to high pressure**



## Spontaneous Processes

### Irreversible

Reversible

Irreversible

Spontaneous processes occur naturally in one way, which are irreversible because they can be reversed only by taking a different path to get back to their original state.

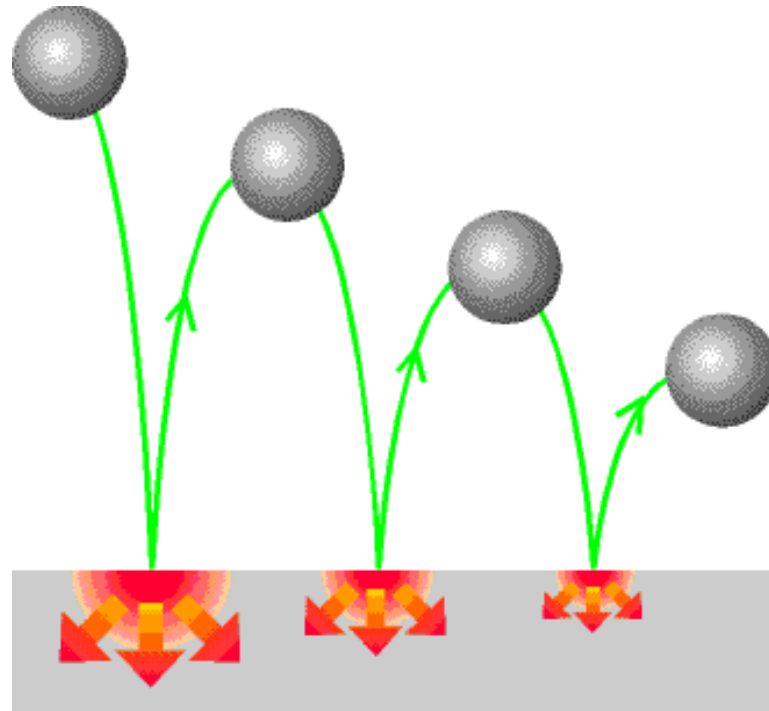


## 3.1 The dispersal of energy

Some things happen naturally. A gas expands to fill the available volume, a hot body cools to the temperature of its surroundings, and a chemical reaction runs in one direction rather than another.

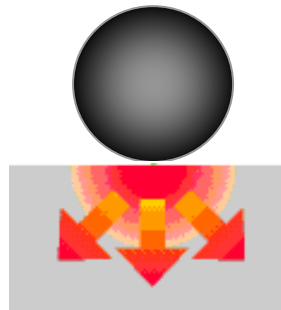
But some things don't. We can confine a gas to a smaller volume, we can cool an object with a refrigerator, and we can force some reactions to go in reverse. However, none of these processes happens spontaneously.

- **Gas: from high pressure to low pressure**
- **Heat: from high temperature to low temperature**
- **Diffusion: from high concentration to low concentration**
- **Reaction: neutralization**

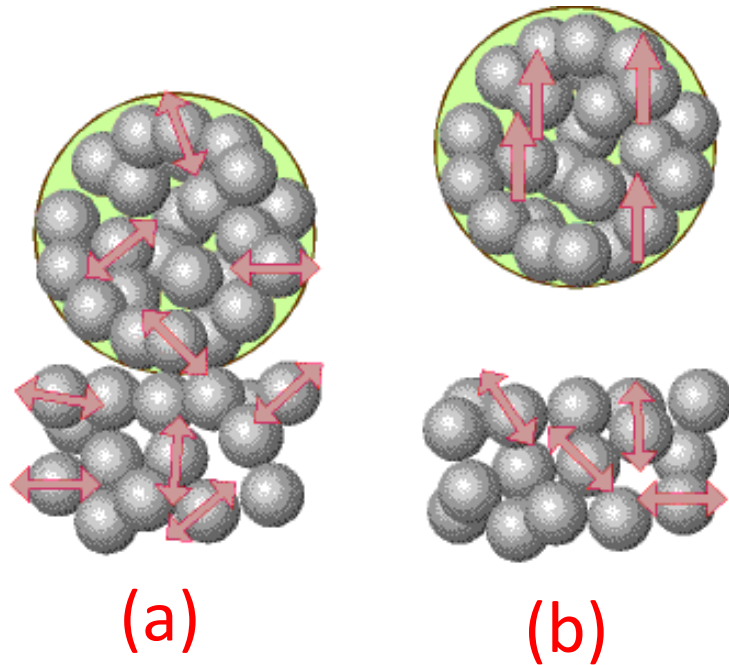


**Sketch for the distribution of energy**

**The spontaneous changes are always accompanied by a dispersal of energy into a more disordered form.**



**Sketch for the distribution of energy**



Sketch for the distribution of energy

**(a) A ball resting on a warm surface; the atoms are undergoing thermal motion;**

**(b) For the ball to fly upwards, some of the random vibrational motion would have to change into coordinated, directed motion.**

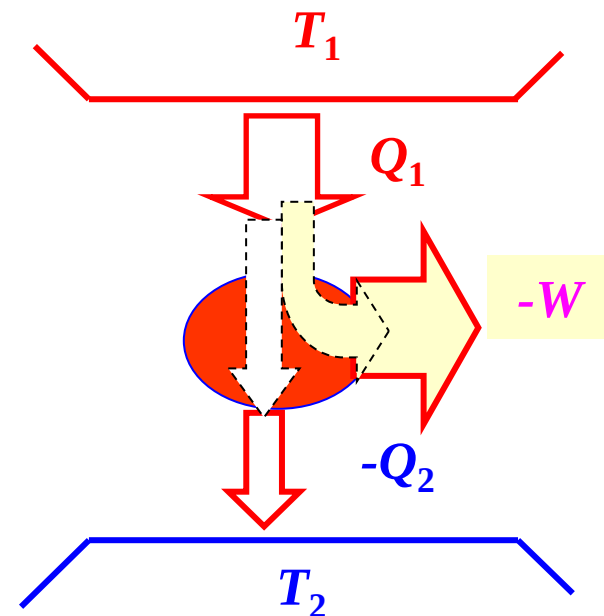
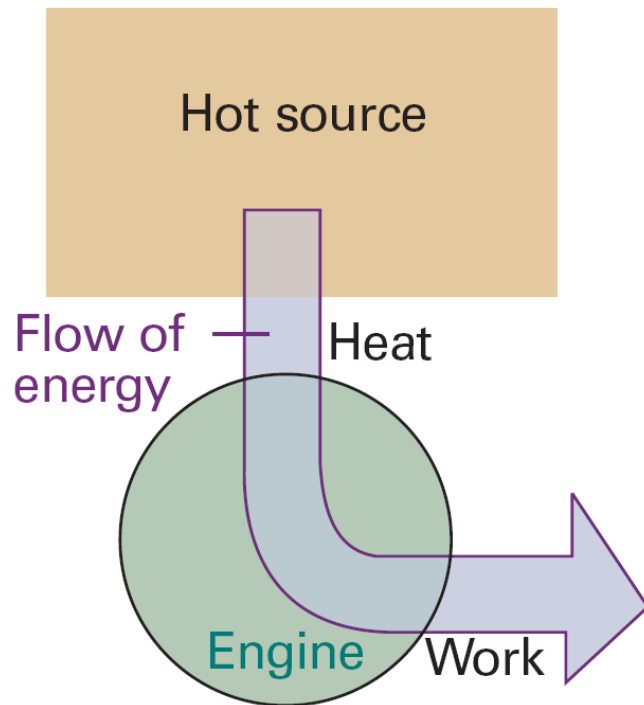


**heat**  **work**

No process is possible in which the sole result is the absorption of heat from a reservoir and its complete conversion into work.



## 3.2 Carnot cycle and Carnot theorem

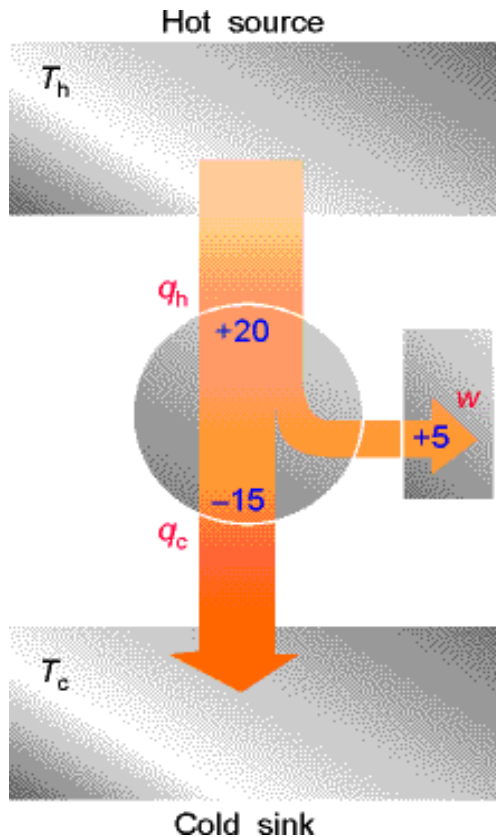


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The efficiency of a heat engine,  $\varepsilon$  :

$$\varepsilon = \frac{\text{work performed}}{\text{heat adsorbed}} = \frac{|w|}{q_h}$$

$$\varepsilon = \frac{q_h + q_c}{q_h} = 1 + \frac{q_c}{q_h} \quad q_c < 0$$

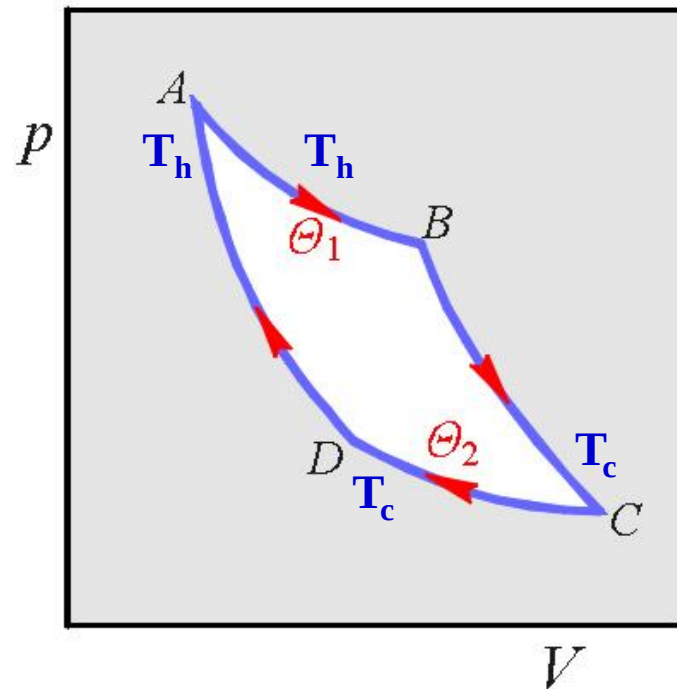


The structure of a heat engine

The efficiency is the work done divided by the heat supplied from the hot source.



- $A \rightarrow B$ : isothermal reversible expansion**
- $B \rightarrow C$ : adiabatic reversible expansion**
- $C \rightarrow D$ : isothermal reversible compression**
- $D \rightarrow A$ : adiabatic reversible compression**





**A→B: isothermal reversible expansion**

For perfect gas,  $U=U(T)$ ,  $q_h+w=0$

$$q_h = -w = nRT_h \ln(V_B/V_A) \quad (1)$$

**B→C: adiabatic reversible expansion**

$$V_C T_c^c = V_B T_h^c \quad (2)$$

**C→D: isothermal reversible compression**

For perfect gas,  $U=U(T)$ ,  $q_c+w=0$

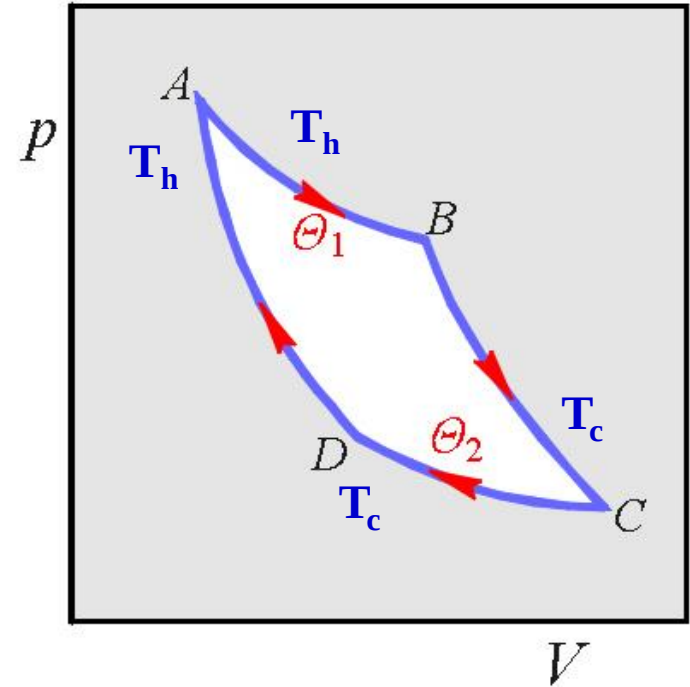
$$q_c = -w = nRT_c \ln(V_D/V_C) \quad (3)$$

**D→A: adiabatic reversible compression**

$$V_A T_h^c = V_D T_c^c \quad (4)$$

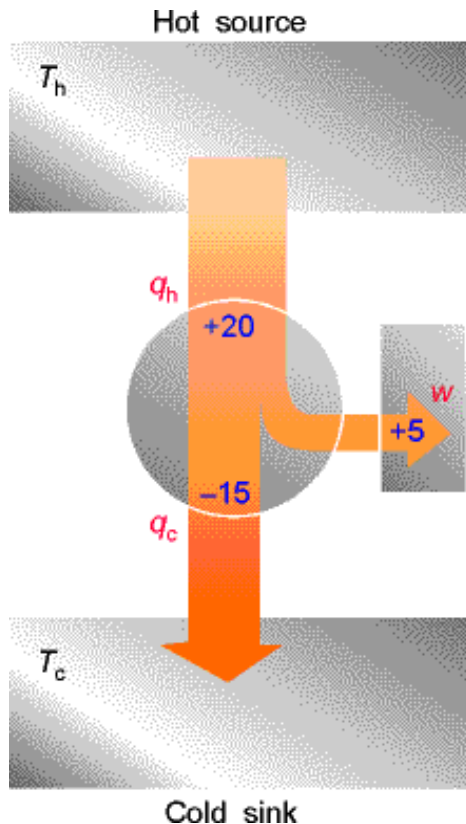
$$(2) \times (4) \longrightarrow V_A V_C T_h^c T_c^c = V_D V_B T_h^c T_c^c$$

$$\longrightarrow V_A V_C = V_D V_B$$



$$\frac{q_h}{q_c} = \frac{nRT_h \ln(V_B/V_A)}{nRT_c \ln(V_D/V_C)} = -\frac{T_h}{T_c}$$

## For a Carnot cycle:



The structure of a heat engine

$$\varepsilon = \frac{q_h + q_c}{q_h} = 1 + \frac{q_c}{q_h} = 1 - \frac{T_c}{T_h}$$

$$\varepsilon_{rev} = 1 - \frac{T_c}{T_h}$$

$$\varepsilon \leq \varepsilon_{rev}$$

The second law of thermodynamics implies that all reversible engines have the same efficiency regardless of their construction.



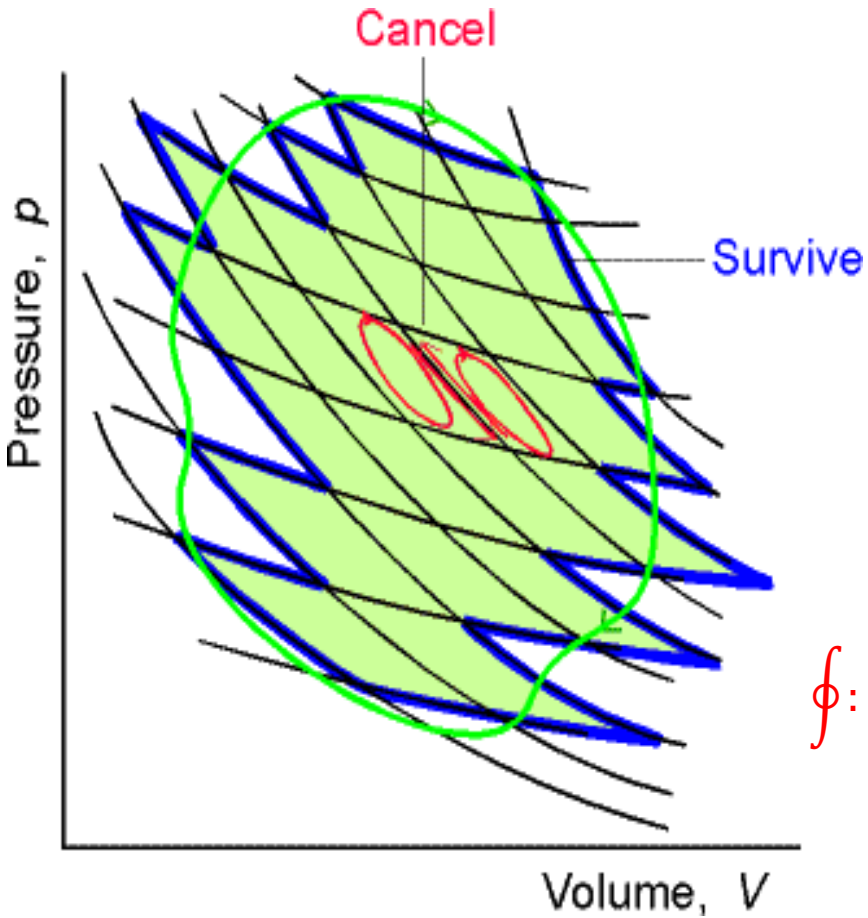
For a reversible circle path

$$\varepsilon_{rev} = 1 + \frac{q_c}{q_h} = 1 - \frac{T_c}{T_h}$$

$$\Rightarrow 1 + \frac{q_c}{q_h} = 1 - \frac{T_c}{T_h}$$

$$\times \frac{q_h}{T_c} \Rightarrow \frac{q_c}{q_h} + \frac{T_c}{T_h} = 0$$

$$\Rightarrow \frac{q_c}{T_c} + \frac{q_h}{T_h} = 0$$



$$\oint \frac{\delta q_{rev}}{T} = 0$$

$dS$

$\oint$ : denotes integration around a closed path



The maximum work can be obtained by operating a reversible engine using the body at  $T$  as the source and the ambient atmosphere at  $T_0$  as the sink.

$$dw = dq \times \varepsilon = dq \times \left(1 - \frac{T_0}{T}\right)$$

$$|w_{max}| = \int_{T_1}^{T_2} dq \times \left(1 - \frac{T_0}{T}\right)$$

$$|w_{max}| = q_{input} - T_0 \int_{T_1}^{T_2} \frac{dq}{T} \Rightarrow dS = \frac{dq_{rev}}{T}$$

Entropy:

A system's thermal energy per unit temperature that is unavailable for doing useful work.

It is a measure of the effectiveness of a specific amount of energy



### 3.3 The thermodynamic definition of entropy

The thermodynamic definition of entropy is based on

$$dS = \frac{dq_{\text{rev}}}{T}$$

The units of entropy:  $\text{J K}^{-1}$ .

The units of Molar entropy:  $\text{J K}^{-1} \text{mol}^{-1}$  (the same as that of  $R$ )



$$dS = \frac{dq_{\text{rev}}}{T}$$

- Entropy is a measure of molecular disorder of a system.
- Heat stimulates disorderly motion in the surroundings. Work, which stimulates uniform motion of atoms in the surroundings, does not change the degree of disorder and so does not change the entropy.
- The change in entropy when a given quantity of heat is transferred will be greater when it is transferred to a cold body than it is transferred to a hot body.





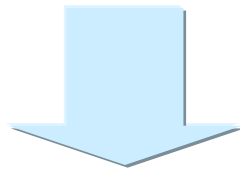
For a measurable change between two states **i** and **f** this expression integrates to:

$$\Delta S = \int_i^f \frac{dq_{\text{rev}}}{T}$$

To calculate the difference in entropy between any two states of a system, integrate the heat supplied at each stage of the path divided by the temperature at which the heat is supplied.

## Example

Calculate the change in entropy when 50kJ of energy is transferred reversibly and isothermally as heat to a large block of copper at (a) 0°C, and (b) 70 °C.



## Example

**Method:**  $\Delta S = \frac{1}{T} \int_i^f dq_{\text{rev}} = \frac{q_{\text{rev}}}{T}$

**Answer:**

$$(a) \quad \Delta S = \frac{q_{\text{rev}}}{T} = \frac{50 \times 10^3 \text{ J}}{273 \text{ K}} = 1.8 \times 10^2 \text{ JK}^{-1}$$

$$(b) \quad \Delta S = \frac{q_{\text{rev}}}{T} = \frac{50 \times 10^3 \text{ J}}{(70 + 273) \text{ K}} = 1.5 \times 10^2 \text{ JK}^{-1}$$



## The change in entropy of the surroundings

$$dS_{\text{sur}} = \frac{dq_{\text{sur,rev}}}{T_{\text{sur}}}$$

Because the temperature and pressure of the surroundings is commonly constant, whatever the change takes place, reversibly or irreversibly, we have

$$dS_{\text{sur}} = \frac{dq_{\text{sur}}}{T_{\text{sur}}}$$
$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T_{\text{sur}}} = \frac{-q}{T_{\text{sur}}}$$



The change of entropy of the surroundings can be calculated by dividing the heat transferred by  $T$  at which the transfer takes place (regardless of how the change is brought about in the system).

For any adiabatic change,

$$q_{\text{sur}} = 0, \text{ so}$$

$$\Delta S_{\text{sur}} = 0$$

## Illustration

To calculate the entropy change in the surroundings when 1.00 mol  $\text{H}_2\text{O}(\text{l})$  is formed from its elements under standard conditions at 298 K, we use  $\Delta H^\ominus = -286 \text{ kJ}$  from a table. The heat released is supplied to the surroundings, now regarded as being at constant pressure, so  $q_{\text{sur}} = +286 \text{ kJ}$ . Therefore,

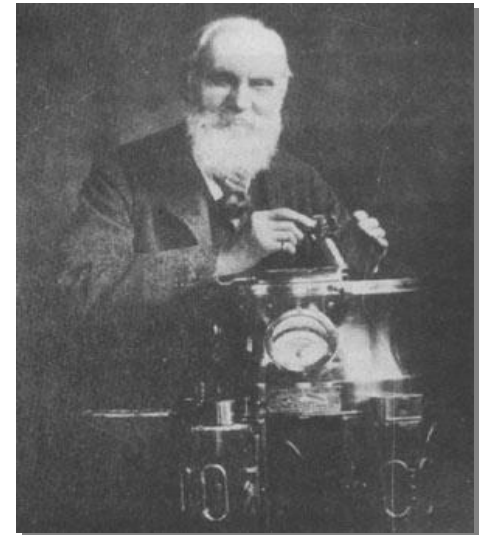
$$\Delta S_{\text{sur}} = \frac{2.86 \times 10^5 \text{ J}}{298 \text{ K}} = +959 \text{ JK}^{-1}$$

## 3.4 The second law



**Clausius:** It is impossible for a self acting machine working in a cyclic process without any external force, to transfer heat from a body at a lower temperature to a body at a higher temperature.

**Kelvin:** No process is possible in which the sole result is the absorption of heat from a reservoir and its complete direction into work.





$$\Delta S = \sum_i \left( \frac{\delta q_i}{T} \right)_{rev}$$

For a reversible cycle path:

$$\Delta S = \oint \left( \frac{\delta q_i}{T} \right)_{rev} = 0$$



For a irreversible process:

$$\varepsilon = 1 + \frac{q_c}{q_h} < \varepsilon_{rev} = 1 - \frac{T_c}{T_h}$$

$$\Rightarrow 1 + \frac{q_c}{q_h} < 1 - \frac{T_c}{T_h} \Rightarrow \frac{q_c}{q_h} + \frac{T_c}{T_h} < 0$$

$$\times \frac{q_h}{T_c} \Rightarrow \frac{q_c}{T_c} + \frac{q_h}{T_h} < 0$$

$$\Rightarrow \oint \left( \frac{\delta q_i}{T} \right)_{ir} < 0$$

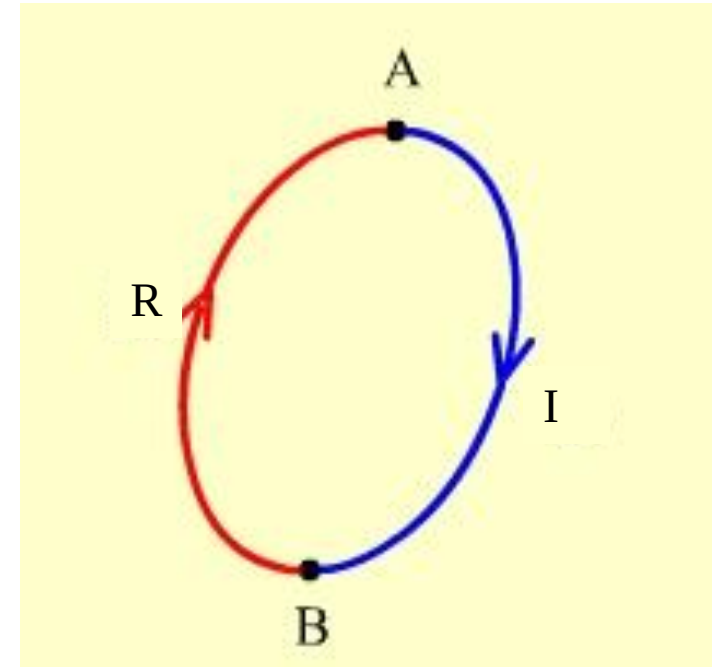
# The Clausius inequality

$$\left( \sum_i \frac{\delta q_i}{T_i} \right)_{I, A \rightarrow B} + \left( \sum_i \frac{\delta q_i}{T_i} \right)_{R, B \rightarrow A} < 0$$

$$\left( \sum_i \frac{\delta q_i}{T_i} \right)_{R, B \rightarrow A} = S_A - S_B$$

$$S_B - S_A > \left( \sum_i \frac{\delta q_i}{T_i} \right)_{I, A \rightarrow B}$$

$$\Delta S_{A \rightarrow B} - \left( \sum_A^B \frac{\delta q}{T} \right) \geq 0$$





$$\Delta S_{A \rightarrow B} \geq \left( \sum_A^B \frac{\delta q}{T} \right)$$

Entropy is a state function

Any process:

$$\Delta S_{A \rightarrow B} = \left( \sum_A^B \frac{\delta q}{T} \right)$$

Reversible

$$\Delta S_{A \rightarrow B} > \left( \sum_A^B \frac{\delta q}{T} \right)$$

Irreversible

$$\Delta S_{A \rightarrow B} < \left( \sum_A^B \frac{\delta q}{T} \right)$$

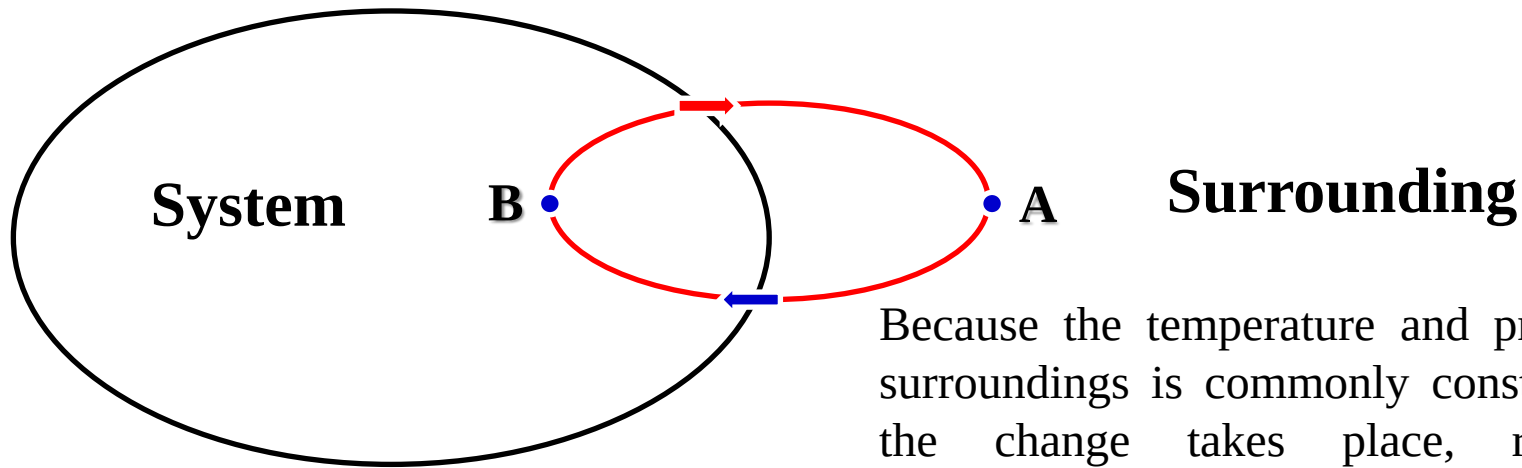
Impossible



For the Adiabatic system:

$$\delta q = 0 \quad \longrightarrow \quad \Delta S \geq 0$$

$$\begin{aligned} \Delta S > 0 & \quad \text{irreversible} \\ \Delta S = 0 & \quad \text{reversible} \end{aligned}$$



Because the temperature and pressure of the surroundings is commonly constant, whatever the change takes place, reversibly or irreversibly, we have:

$$\Delta S_{A \rightarrow B} - \left( \sum_A^B \frac{\delta q}{T} \right) \geq 0$$

$\Delta S_{sys}$

$\Delta S_{sur} = \int \frac{dq_{rev,sur}}{T} = \int \frac{dH_{sur}}{T_{sur}} = -\frac{q}{T_{sur}}$



**For the isolated system:**

$$\Delta S_{iso} = \Delta S + \Delta S_{sur} \geq 0$$

**In an isolated system the entropy of the system alone cannot decrease when a spontaneous change takes place.**

$$\Delta S \geq 0$$

**> spontaneous**

**= reversible**



## The principle of increase of entropy

**The entropy of an isolated system increases in the course of a spontaneous change:**

**Thermodynamically irreversible processes are spontaneous processes, and must be accompanied by an increase in entropy.**



- 孤立系统中，自发发生的方向永远朝着熵增加的方向。
- 熵可以判断过程的方向性，但只能判断孤立系统。

$$\Delta S_{\text{孤}} = \Delta S + \Delta S_{\text{环}} \geq 0$$

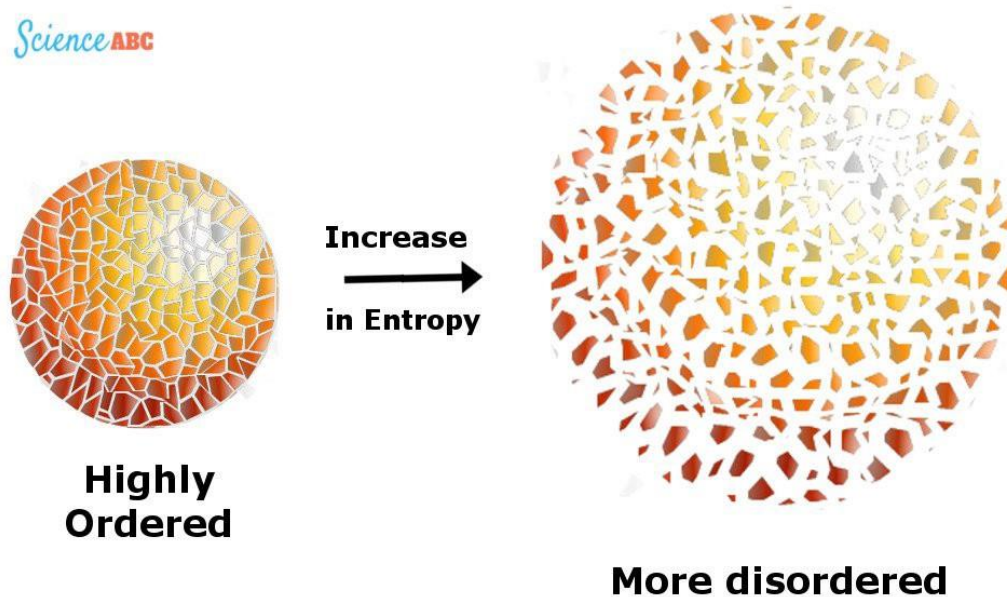
熵的物理意义：

- ① 在孤立系统中，熵增加=能量贬值（能量的可用性降低）
- ② 熵是孤立系统中能量不可用性的量度，熵越大，能量可用性越差。

热力学第二定律物理意义：

自然界实际过程向着能量可用性减小的方向进行（能量贬值）

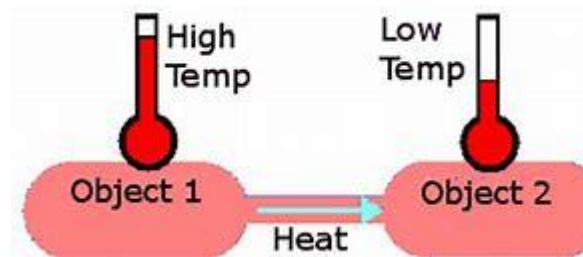
## Entropy: A criteria for spontaneity



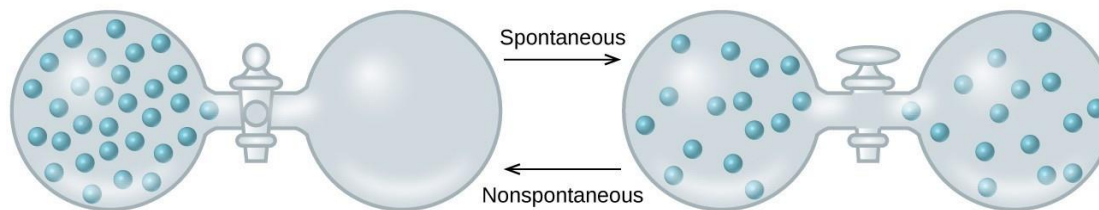
A measure of the level of disorder of a system

A measure of the unavailability of a specific amount of energy

**Spontaneous processes have a natural tendency to occur in one direction under a given set of conditions.**



**Heat: from high temperature to low temperature**



**Gas: from high pressure to low pressure**

# The End of Universe



Heat death theory is the most probable way our Universe would end. It would keep on expanding and losing heat till it reaches absolute zero.



# CONCEPT VIGNETTES

A Collaboration between SUTD and MIT







## 3.5 Entropy changes accompanying specific processes

**Because a change in the degree of molecular order occurs when a substance freezes or boils, we should expect the transition to be accompanied by a change in entropy.**

**In a reversible process**

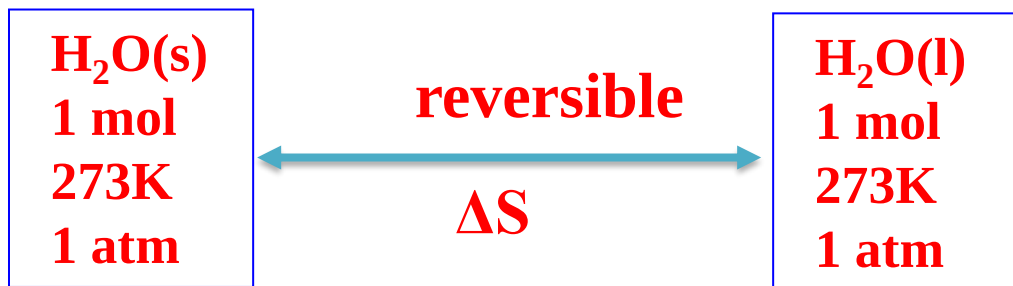
**In an irreversible process**

## A. The entropy of phase transition at the transition temperature

At the normal transition temperature,  $T_{\text{trs}}$ , any transfer of heat between the system and its surroundings is **reversible** because the two phases in the system are in equilibrium.

Since  $q = \Delta_{\text{trs}} H$  at constant  $p$ , the change in molar entropy of the system is

$$\Delta_{\text{trs}} S = \frac{\Delta_{\text{trs}} H}{T_{\text{trs}}}$$

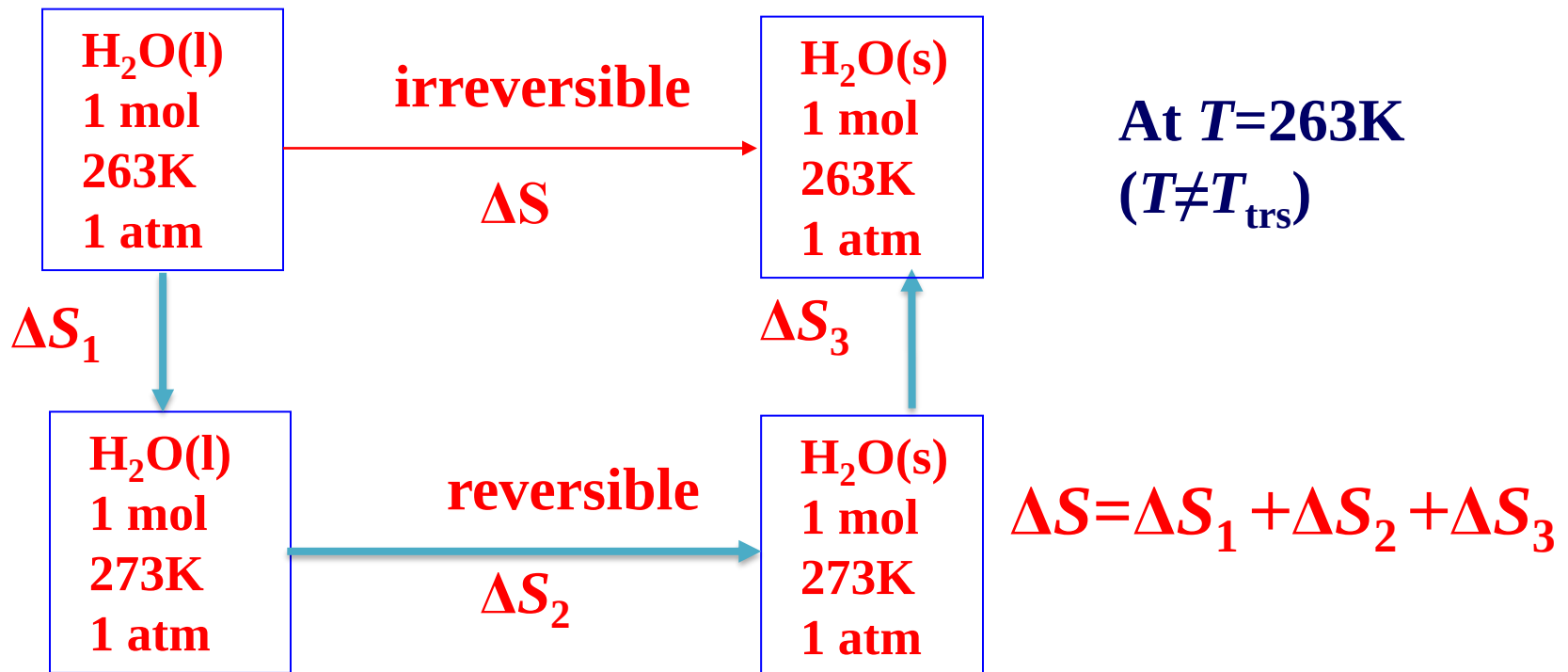


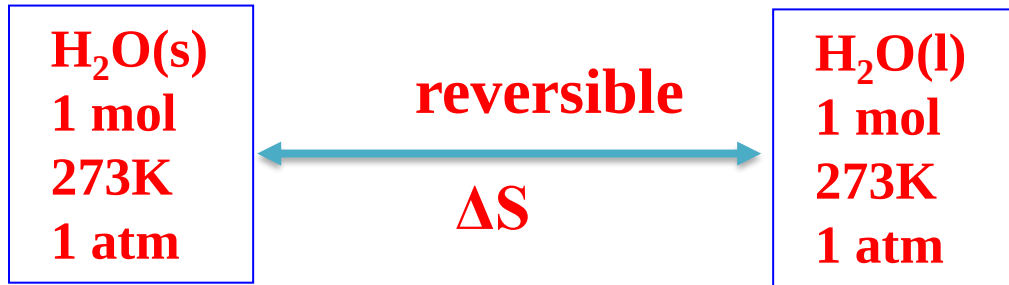
At  $T_{\text{trs}} = 273\text{K}$

$$\Delta_{\text{trs}} S = \frac{\Delta_{\text{trs}} H}{T_{\text{trs}}}$$



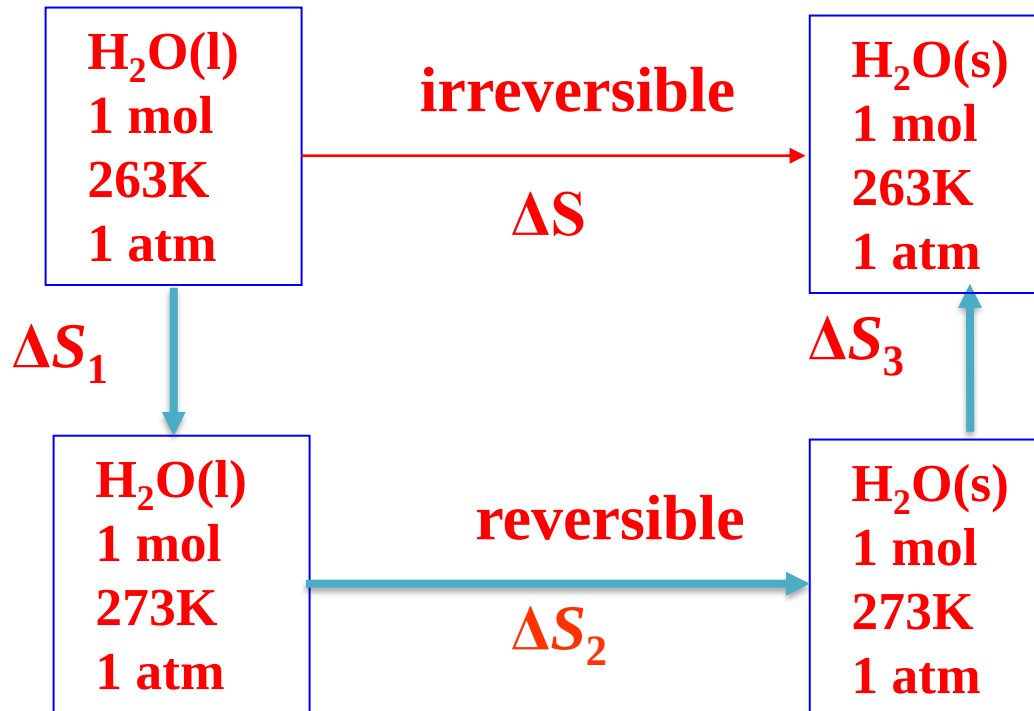
If a process changes at which the temperature deviates the normal transition temperature,  $T_{\text{trs}}$ , the process is not a reversible and the above equation can not be directly used to calculate the change in entropy.





At  $T_{\text{trs}} = 273\text{K}$

$$\Delta_{\text{trs}} S = \frac{\Delta_{\text{trs}} H}{T_{\text{trs}}}$$



At  $T = 263\text{K}$   
( $T \neq T_{\text{trs}}$ )

$$\Delta S = \Delta S_1 + \Delta S_2 + \Delta S_3$$



If the phase transition is **exothermic** ( $\Delta_{\text{trs}}H < 0$ ), the entropy change is negative,  $\Delta_{\text{trs}}S < 0$ , which is consistent with the system becoming more ordered.

If the transition is **endothermic** ( $\Delta_{\text{trs}}H > 0$ ), the entropy change is positive,  $\Delta_{\text{trs}}S > 0$ , which is consistent with the system becoming more disordered.



## B. The isothermal reversible expansion of a perfect gas

The change in entropy of a perfect gas that expands isothermally from  $V_i$  to  $V_f$  is

$$\Delta S = \frac{1}{T} \int_i^f dq_{\text{rev}} = \frac{q_{\text{rev}}}{T}$$

The heat absorbed during a reversible isothermal expansion of a perfect gas can be calculated from:

$$\Delta U = q + w \text{ and } \Delta U = 0.$$

For a reversible change  $q_{\text{rev}} = -w_{\text{rev}}$



$$w_{\text{rev},T} = -nRT \int_{V_i}^{V_f} \frac{dV}{V} = -nRT \ln \frac{V_f}{V_i}$$

$$q_{\text{rev}} = -w_{\text{rev}} = nRT \ln \frac{V_f}{V_i} \text{ It follows that}$$

**The change in entropy of a perfect gas that expands isothermally from  $V_i$  to  $V_f$  is**

$$\underline{\Delta S = nR \ln \frac{V_f}{V_i}}$$



$$\Delta S = nR \ln \frac{V_f}{V_i}$$

- Because  $S$  is a state function, this expression applies whether the change of state occurs reversibly or irreversibly.
- The total change in entropy does depend on how the expansion takes place
  - If the change is reversible, the entropy change in the surroundings must be such as to give  $\Delta S_{\text{tot}} = 0$ . Therefore, in this case,  $\Delta S_{\text{sur}}$  must be negative.
  - If the expansion occurs freely ( $w = 0$ ) and irreversibly, and if  $T$  remains constant, then  $q = 0$ . Consequently,  $\Delta S_{\text{sur}} = 0$ , and the total entropy change is given by this equation.

## C. The variation of entropy with temperature

The entropy of a system at a temperature  $T_f$  from a knowledge of its entropy at a temperature  $T_i$  and the heat supplied to change its temperature from one value to the other:

$$\Delta S = S(T_f) - S(T_i) \quad \downarrow$$
$$S(T_f) = S(T_i) + \int_i^f \frac{dq_{\text{rev}}}{T}$$

## 等压变温

At constant pressure, from the definition of **constant-pressure heat capacity**, so long as the system is doing no non-expansion work:

$$\begin{aligned}dq_{\text{rev}} &= C_p dT \\ S(T_f) &= S(T_i) + \int_i^f \frac{dq_{\text{rev}}}{T} \\ &= S(T_i) + \int_i^f \frac{C_p dT}{T}\end{aligned}$$

When  $C_p$  is independent of  $T$  in the temperature range of interest:

$$S(T_f) = S(T_i) + C_p \ln\left(\frac{T_f}{T_i}\right)$$

## 等容变温

**With constant volume, from the definition of constant-volume heat capacity:**

$$\begin{aligned}dq_{\text{rev}} &= C_V dT \\ S(T_f) &= S(T_i) + \int_i^f \frac{dq_{\text{rev}}}{T} \\ &= S(T_i) + \int_i^f \frac{C_V dT}{T}\end{aligned}$$

**When  $C_V$  is independent of  $T$  in the temperature range of interest:**

$$S(T_f) = S(T_i) + C_V \ln\left(\frac{T_f}{T_i}\right)$$



## 相平衡

$$\Delta_{trs}S = \frac{\Delta_{trs}H}{T_{trs}}$$

## 恒温可逆膨胀

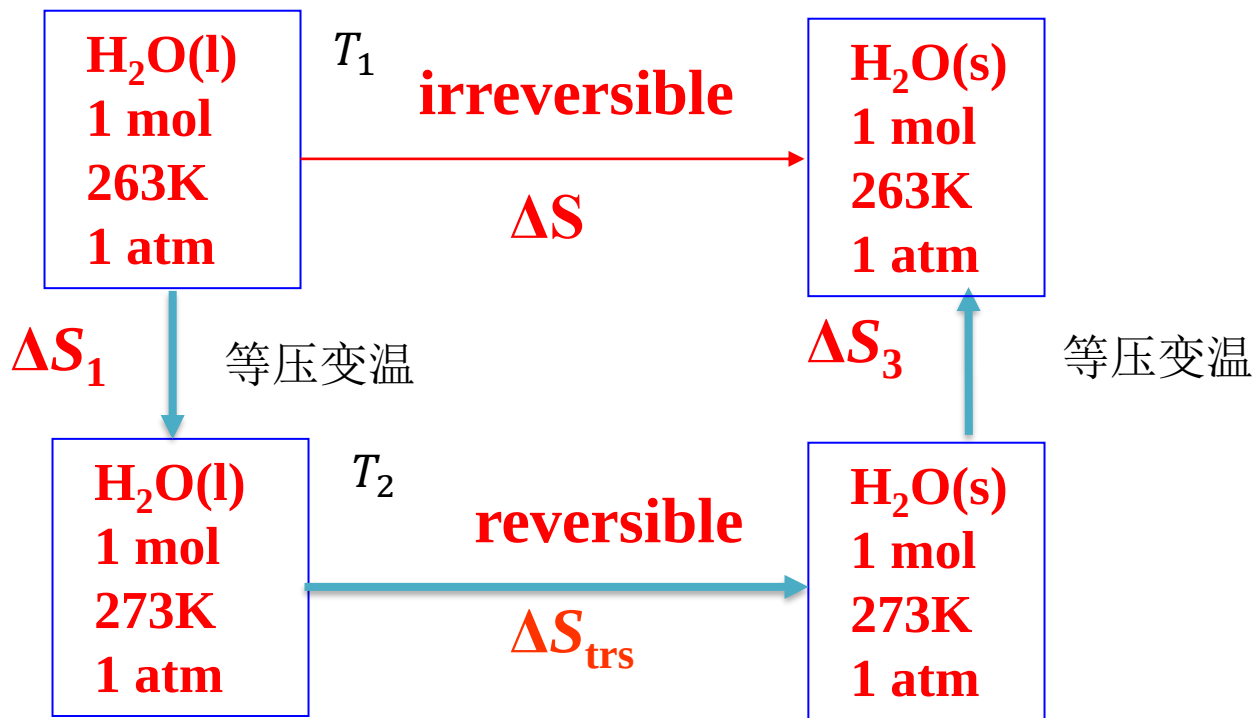
$$\Delta S = nR \ln \frac{V_f}{V_i}$$

## 等压变温

$$\Delta S = C_p \ln \frac{T_f}{T_i}$$

## 等容变温

$$\Delta S = C_V \ln \frac{T_f}{T_i}$$



$$\Delta S = \Delta S_1 + \Delta S_{trs} + \Delta S_3$$

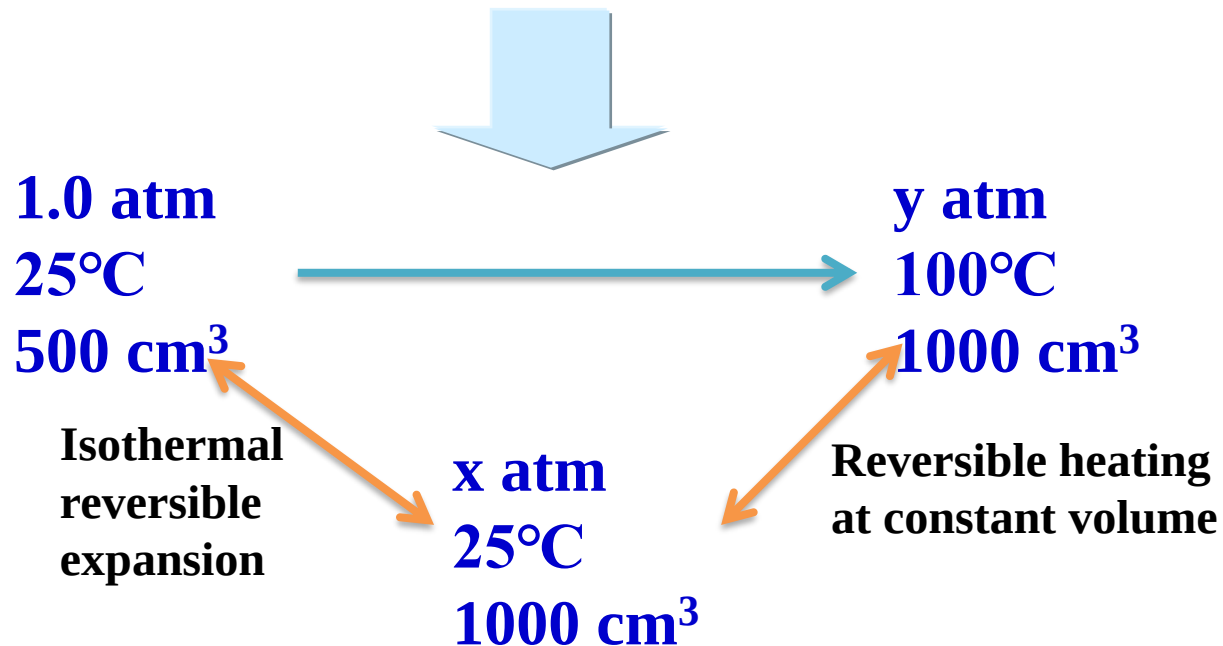
$$= C_p(l) \ln \frac{T_2}{T_1} + \frac{\Delta_{trs} H}{T_2} + C_p(g) \ln \frac{T_1}{T_2}$$

↓

$$T_{trs}$$

## Example - Calculating the entropy change

Calculate the entropy change when argon at  $25^{\circ}\text{C}$  and  $1.00\text{ atm}$  in a container of volume  $500\text{ cm}^3$  is allowed to expand to  $1000\text{ cm}^3$  and is simultaneously heated to  $100^{\circ}\text{C}$ .



## Example - Calculating the entropy change

**Method:** Because  $S$  is a state function, we are free to choose the most convenient path from the initial state.

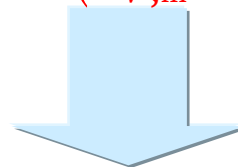
**The first step:** reversible isothermal expansion to the final volume. The entropy change in this step is

$$\Delta S_1 = nR \ln(V_f / V_i)$$

**The second step:** reversible heating at constant volume to the final temperature. The entropy change in this step is

$$\Delta S_2 = C_{V,m} \ln(T_f / T_i) \quad (C_{V,m} = C_{p,m} - R)$$

$$\Delta S = \Delta S_1 + \Delta S_2$$



## Example - Calculating the entropy change

**Answer:** The amount of Ar present is (from the perfect gas equation and the data for the initial state):

$$n = pV/RT = 0.0204 \text{ mol}$$

$$\begin{aligned}\Delta S_1 &= nR \ln(V_f/V_i) \\ &= nR \ln(1000/500) = 0.118 \text{ J K}^{-1}\end{aligned}$$

From  $\Delta S = C_p \ln(T_f/T_i)$  and  $C_{p,m} - C_{V,m} = R$

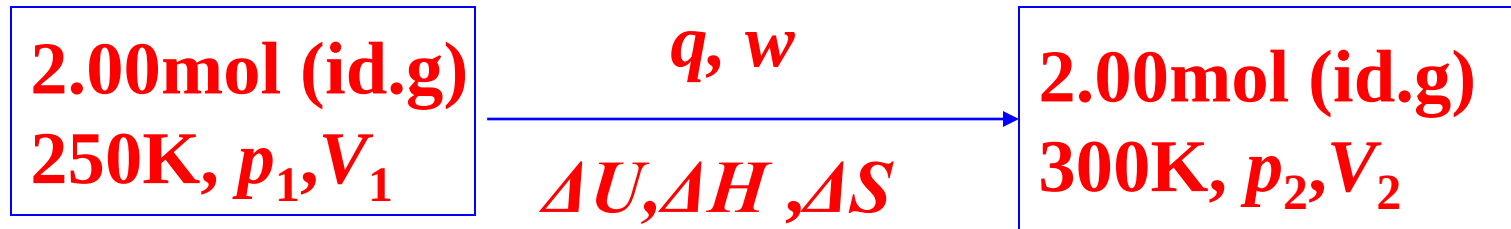
$$\begin{aligned}\Delta S_2 &= (0.0204 \text{ mol}) \times (12.48 \text{ J K}^{-1} \text{ mol}^{-1}) \times \ln(373 \text{ K}/298 \text{ K}) \\ &= 0.057 \text{ J K}^{-1}\end{aligned}$$

$$\Delta S = \Delta S_1 + \Delta S_2 = 0.118 \text{ J K}^{-1} + 0.057 \text{ J K}^{-1} = 0.175 \text{ J K}^{-1}$$

## Example

- A sample consisting of 2.00 mol of diatomic perfect gas at 250K is compressed reversibly and adiabatically until its temperature reaches 300K. Calculate  $q$ ,  $w$ ,  $\Delta U$ ,  $\Delta H$  and  $\Delta S$ . Given that  $C_{V,m} = 27.5 \text{ K}^{-1} \text{ mol}^{-1}$ .

# Example



**Answer:** Because the compression process is reversible and adiabatic, it follows that

$$q = q_{\text{rev}} = 0$$

$$\Delta S = \int_i^f \frac{dq_{\text{rev}}}{T} = 0$$

## Example

$$\begin{aligned}\Delta U &= nC_{V,m}\Delta T \\ &= (2.0\text{mol}) \times (27.5\text{JK}^{-1}\text{mol}^{-1}) \times (300 - 250)\text{K} \\ &= 2750\text{J} = +2.75\text{kJ}\end{aligned}$$

$$w = \Delta U - q = \Delta U = 2.75\text{ kJ}$$

Since  $\Delta H = nC_{p,m}\Delta T$  and

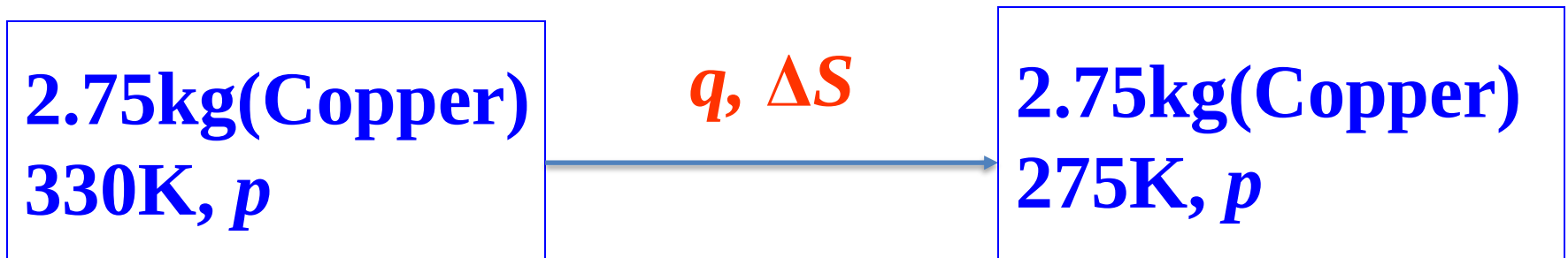
$$C_{p,m} = C_{V,m} + R = (27.5 + 8.314)\text{JK}^{-1}\text{mol}^{-1} = 35.814\text{JK}^{-1}\text{mol}^{-1}$$

then

$$\begin{aligned}\Delta H &= (2.0\text{mol}) \times (35.814\text{JK}^{-1}\text{mol}^{-1}) \times (300 - 250)\text{K} \\ &= 3581.4\text{J} = 3.58\text{kJ}\end{aligned}$$

## Example

A sample of copper of mass 2.75 kg is cooled at constant pressure from 330K to 275K. Calculate (a) the energy that must be removed as heat, and (b) the change in entropy of the sample.



## Example

**Answer:** The heat released is

$$q = C_p \Delta T = n C_{p,m} \Delta T$$

$$= \left( \frac{2.75 \text{ kg}}{63.54 \times 10^{-3} \text{ kg mol}^{-1}} \right) \times (24.44 \text{ J K}^{-1} \text{ mol}^{-1}) \times (275 - 330) \text{ K}$$

$$= -58.2 \times 10^3 \text{ J}$$

## Example

(b)

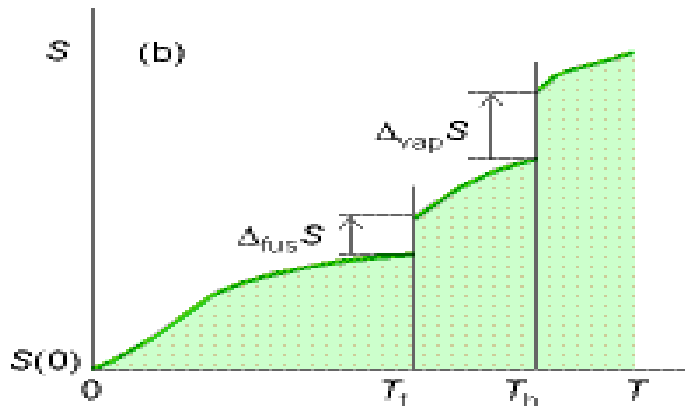
$$\Delta S = \int \frac{d\mathbf{q}_{\text{rev}}}{T} = \int \frac{C_p dT}{T} = nC_{p,m} \ln \frac{T_f}{T_i}$$

$$= \left( \frac{2.75 \text{ kg}}{63.54 \times 10^{-3} \text{ kg mol}^{-1}} \right) \times (24.44 \text{ JK}^{-1} \text{ mol}^{-1}) \times \ln \frac{275}{330}$$

$$= -193 \text{ JK}^{-1}$$

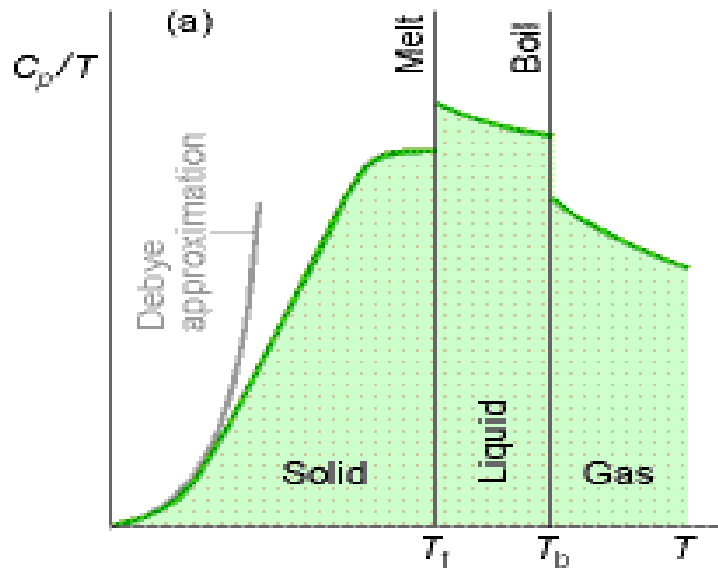
## D. The measurement of entropy

**The entropy of a system at a temperature  $T$  based on its entropy at  $T = 0$**



$$S(T) = S(0) + \int_0^{T_f} \frac{C_p(\text{s})dT}{T} + \frac{\Delta_{\text{fus}} H}{T_f} + \int_{T_f}^{T_b} \frac{C_p(\text{l})dT}{T} + \frac{\Delta_{\text{vap}} H}{T_b} + \int_{T_b}^T \frac{C_p(\text{g})dT}{T}$$

**The entropy is equal to the area beneath the upper curve up to the corresponding temperature, plus the entropy of each phase transition passed.**



One problem with the determination of entropy is the difficulty of measuring heat capacities near  $T=0$ . There are good theoretical grounds for assuming that the heat capacity is proportional to  $T^3$  when  $T$  is low, and this dependence is the basis of the Debye extrapolation.

**The variation of  $C_p/T$  with the temperature**

## Debye extrapolation

Theoretically, the heat capacity is proportional to  $T^3$  when  $T$  is low:

$$C_p = aT^3$$



## 3.6 The third law of thermodynamics

### A. Nernst heat theorem

**The entropy change accompanying any physical or chemical transformation approaches zero as the temperature approaches zero:**

$$\Delta S_{T \rightarrow 0 \text{ K}} = 0$$

## B. Third law of thermodynamics

If the entropy of every element in its most stable state at  $T=0$  is taken as zero, then every substance has a positive entropy which at  $T=0$  may become zero, and which does become zero for all perfect crystalline substances, including compounds.

**$S(0K) = 0$ , for all perfect crystalline substances**

The Third Law **does not state** that entropies are zero at  $T = 0$ ; it merely implies that all perfect materials have the same entropy at that temperature. Choosing this common value as zero is then a matter of convenience.



## C. Third-Law entropies

**The Third-Law entropy, standard entropy,  $S^{\ominus}(T)$ :**

**On the basis of  $S(0\text{K}) = 0$  for all perfect crystals, the entropy of the substance in its standard state at the Temperature  $T$  is denoted  $S^{\ominus}(T)$**



**The standard reaction entropy,  $\Delta_r S^\ominus$** : the difference between the molar entropies of the pure, separated products and the pure, separated reactants, all substances being in their standard states at the specified temperature.

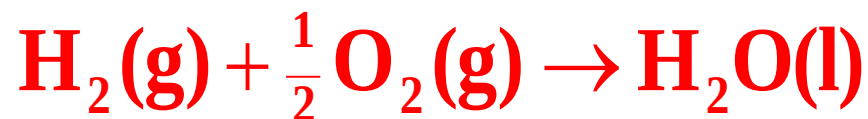
$$\Delta_r S^\ominus = \sum_{\text{Products}} \nu S_m^\ominus - \sum_{\text{Reactants}} \nu S_m^\ominus$$

$$\Delta_r S^\ominus = \sum_J \nu_J S_m^\ominus \text{ (J)}$$

**$\nu$ -stoichiometric number**

## Illustration

Calculate the standard reaction entropy of the following reaction at 25 °C :



$$\Delta_r S^\ominus = \sum_{\text{Products}} \nu S_{\text{m}}^\ominus - \sum_{\text{Reactants}} \nu S_{\text{m}}^\ominus$$

$$\Delta_r S^\ominus = S_{\text{m}}^\ominus(\text{H}_2\text{O}, \text{l}) - \left\{ S_{\text{m}}^\ominus(\text{H}_2, \text{g}) + \frac{1}{2} S_{\text{m}}^\ominus(\text{O}_2, \text{g}) \right\}$$

$$= 69.9 - \left\{ 130.7 + \frac{1}{2}(205.0) \right\} \text{JK}^{-1}\text{mol}^{-1} = -163.3 \text{JK}^{-1}\text{mol}^{-1}$$



## Concentrating on the system

Entropy is the basic concept for discussing the direction of natural change, but to use it we have to analyze changes in both the system and its surroundings. We have seen that it is always very simple to calculate the entropy change in the surroundings, and we shall now see that it is possible to devise a simple method for taking that contribution into account automatically. This approach focuses our attention on the system and simplifies discussions.

## 3.7 The Helmholtz and Gibbs energies

For a system in thermal equilibrium with its surroundings at a temperature  $T$ , when a change in the system occurs and there is a transfer of energy as heat between the system and the surroundings, the Clausius inequality is

$$dS - \frac{dq}{T} \geq 0$$

$dS_{sur} = -dq/T$ , where  $dq$  is the heat supplied to the system during the process  
 $dq_{sur} = -dq$ , because the heat that enters the surroundings comes from the system

This inequality can be developed in two ways according to the conditions of (a) constant volume, and (b) constant pressure under which the process occurs

## A. Criteria for spontaneity

Heat transfer at **constant volume**, zero of non-expansion work:

$$dq_V = dU$$

$$dS - \frac{dU}{T} \geq 0$$

$$TdS \geq dU$$

- The criterion for spontaneous change is expressed solely in terms of the state functions of the system.
- At either constant internal energy ( $dU = 0$ ) or constant entropy ( $dS = 0$ ):

$$dS_{U,V} \geq 0 \quad dU_{S,V} \leq 0$$



$$dS_{U,V} \geq 0 \quad dU_{S,V} \leq 0$$

- The first inequality states that, in a system at constant  $V$  and constant  $U$ , the entropy increases in a spontaneous change.
- The second inequality should be interpreted that, if the entropy of the system is unchanged, then there must be an increase in entropy of the surroundings, which can be achieved only if the energy of the system decreases as energy flows out as heat.

Heat transfer at constant pressure, zero of non-expansion work:

$$dq_p = dH$$

$$**TdS \geq dH** \text{ (constant } p, \text{ no non-expansion work)}$$

At either constant enthalpy ( $dH = 0$ ) or constant entropy ( $dS = 0$ ):

$$\mathbf{dS}_{H,p} \geq 0 \qquad \mathbf{dH}_{S,p} \leq 0$$

The entropy of the system at constant  $p$  must increase if its enthalpy remains constant; the enthalpy must decrease if the entropy of the system is constant.



## B. The definition and some remarks on the Helmholtz energy

At constant volume and temperature

$$dS - \frac{q}{T} \geq 0$$

$$TdS - q \geq 0$$

$$TdS - (dU - w) \geq 0$$

$$dU - TdS \leq w$$

**w: non-expansion work**



## Helmholtz function

$$dA = dU - TdS$$

$$\Delta A \leq w$$

Therefore the maximum energy that can be obtained from the system as work, is given by:

$$dw_{\max} = dA$$

$$w_{\max} = \Delta A$$

The change in the Helmholtz energy is equal to the maximum work accompanying a process.  $A$  is also called the ‘maximum work function’, or the ‘work function’.



Heat transfer at **constant volume and temperature**  
 $w = 0$ , no non-expansion work,



$$\Delta A_{T,V} \leq 0$$

**The criterion for spontaneous change is then:**

$$\Delta A_{T,V} \leq 0 \quad \left\{ \begin{array}{l} < 0 \text{ spontaneous} \\ = 0 \text{ equilibriums} \end{array} \right.$$



## The criterion of equilibrium

$$dA_{T,V} \leq 0$$

$$dA = dU - TdS$$

- Systems move spontaneously toward states of lower  $A$  if a path is available
- The system is in equilibrium when neither the forward nor reverse process has a tendency to occur
- The form of  $dA$  may give the impression that systems favor lower energy, but that is misleading
  - $dS$  is the entropy change of the system
  - $-dU/T$  is the entropy change of the surroundings

## Maximum work

The change in the Helmholtz function is equal to the maximum work accompanying a process :

$$dw_{\max} = dA$$

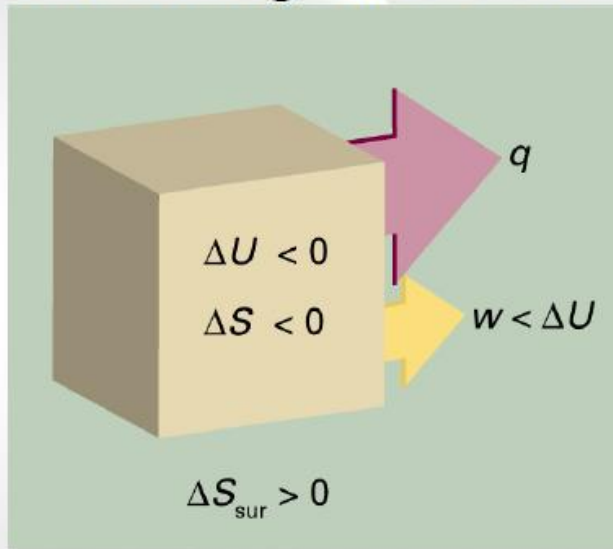
$$w_{\max} = \Delta A$$

$$\Delta A = \Delta U - T\Delta S$$

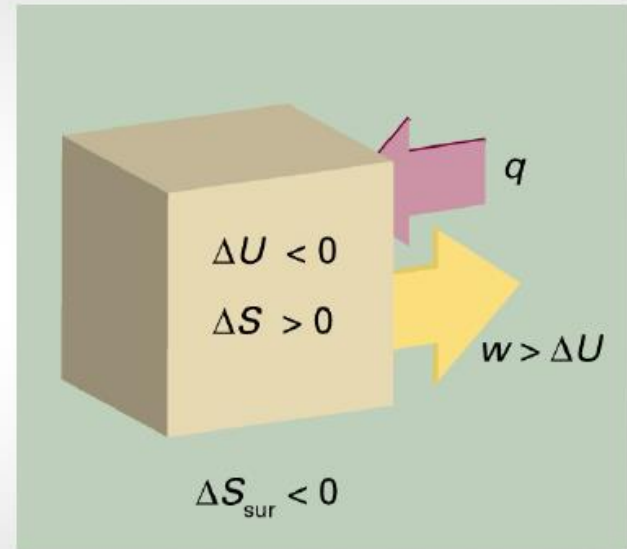
- **$A = U - TS$** :  $A$  is the total internal energy of the system,  $U$ , less a contribution that is stored as energy of thermal motion (the quantity  $TS$ ).
- Because energy stored in random thermal motion cannot be used to achieve uniform motion in the surroundings, only the part of  $U$  that is not stored in that way, the quantity  $U - TS$ , is available for conversion into work.



The process is spontaneous if overall the entropy of the system and its surroundings increase.



The entropy of the system decreases, so that of the surroundings must increase, which means that energy must pass from to the surrounding as heat



The entropy of the system increases; we can afford to lose some entropy of the surroundings. Some of their energy may be lost as heat to the system. This energy can be returned to them as work. The work done can exceed  $\Delta U$

## C. The definition and some remarks on the Gibbs energy

$$dS - \frac{q}{T} \geq 0$$

$$TdS - q \geq 0$$

$$\text{Constant } T, p, \quad q = \Delta H$$

$$TdS - (dH) \geq 0$$

$$dH - TdS \leq 0$$

$$\text{The Gibbs energy, } G: \quad G = H - TS$$

$$\text{At constant } T, p, \quad dG = dH - TdS$$



From the definition that  $H = U + pV$

$$dH = dq + dw + d(pV)$$

For a reversible change

$$dw = dw_{\text{rev}} \quad \text{and} \quad dq = dq_{\text{rev}} = TdS$$

$$\text{so } dG = dH - TdS$$

$$= \cancel{TdS} + dw_{\text{rev}} + d(pV) - \cancel{TdS}$$

$$= dw_{\text{rev}} + d(pV)$$



$dw_{\text{rev}}$

consists of expansion work ( $-pdV$ )  
and non-expansion work ( $dw_{\text{add}}$ )



$$dG = dw_{\text{rev}} + d(pV)$$

$$\begin{aligned} dG &= (-pdV + dw_{\text{e,rev}}) + pdV + Vdp \\ &= dw_{\text{e,rev}} + Vdp \end{aligned}$$

For the reversible change at constant  $p$  and  $T$

$$dG_{\text{rev}} = dw_{\text{add,rev}} \quad \text{non-expansion work}$$



Because the process is reversible, the work done must now have its maximum value.

$$dw_{add,max} = dG_{rev}$$

$$w_{add,max} = \Delta G_{rev}$$

For a reversible process at constant temperature and pressure the **maximum non-expansion work** is equal to the change in **Gibbs energy**.



At constant temperature and pressure:

$$\Delta G \leq w_{add}$$

= *Reversible*

< *Spontaneous*



Heat transfer at **constant pressure and temperature**, no non-expansion work,



$$\Delta G \leq 0$$

**The criterion for spontaneous change is then:**

$$dG_{T,p} \leq 0 \quad \left\{ \begin{array}{l} < 0 \text{ spontaneous} \\ = 0 \text{ equilibriums} \end{array} \right.$$



$$dG_{T,p} \leq 0$$

- Chemists are usually more interested in changes occurring at constant P than that at constant V.
- At constant T and P, chemical reactions are spontaneous in the directions of decreasing Gibbs energy.
- If G decreases as the reaction proceeds, the reaction has a spontaneous tendency to convert reactants into products
- If G increases as the reaction proceeds, the reverse reaction is spontaneous



$$\mathbf{dG} = \mathbf{dH} - T\mathbf{dS}$$

- Endothermic reactions are driven by the increase of entropy of the system
- $dH > 0$  ; In order for the process to be spontaneous, the entropy of the system increases so much that  $TdS$  outweighs  $dH$ .
- This entropy change overcomes the reduction of entropy brought about in the surroundings by the inflow of heat into the system ( $dS_{sur} = -dH/T$  at constant P).

## **Example** - Calculating the maximum available work

**When 1mol C<sub>6</sub>H<sub>12</sub>O<sub>6</sub> (glucose) is oxidized to carbon dioxide and water at 25 °C according to the equation**



**How much of this energy change can be extracted as (a) heat at constant pressure, (b) work? Given that**

$$\Delta_r U^\ominus = -2808 \text{ kJ mol}^{-1}, \Delta_r S = +182.4 \text{ J K}^{-1} \text{ mol}^{-1}$$



The enthalpy of a perfect gas is related to its internal energy by using  $pV = nRT$  in the definition of  $H$ :

$$**$H = U + pV = U + nRT$**$$

This relation implies that the change of enthalpy in reaction that produces or consumes gas is:

$$**$\Delta_r H = \Delta_r U + \Delta n_g RT$**$$

## **Example** - Calculating the maximum available work

**Method:** We know that the heat released at constant pressure is equal to the value of  $\Delta H$ , so we need to relate  $\Delta_r H^\ominus$  to  $\Delta_r U^\ominus$ , which is given. Suppose that all the gases involved are perfect and the equation is

$$\Delta_r H = \Delta_r U + \Delta n_g RT$$

For the maximum available from the process,

$$w_{\max} = \Delta A$$

## **Example** - Calculating the maximum available work

**Answer:** (a) Because  $\Delta n_g = 0$ , so

$$\Delta_r H^\ominus = \Delta_r U^\ominus = -2808 \text{ kJ mol}^{-1}$$

Therefore, at constant pressure, the energy available as heat is  $2808 \text{ kJ mol}^{-1}$ .

(b) Because  $T = 298 \text{ K}$ , the value of  $\Delta_r A$

$$\Delta_r A^\ominus = \Delta_r U^\ominus - T\Delta_r S^\ominus = -2862 \text{ kJ mol}^{-1}$$

Therefore, the combustion of  $1.000 \text{ mol C}_6\text{H}_{12}\text{O}_6$  can be used to produce up to  $2862 \text{ kJ}$  of work.

**Example - Calculating the maximum non-expansion work of a reaction**

**How much energy is available for sustaining muscular and nervous activity from the combustion of 1.00 mol of glucose molecules under standard conditions at 37°C (blood temperature)?**

**Given that The standard entropy of reaction is +182.4 JK<sup>-1</sup> mol<sup>-1</sup> and the  $\Delta_r H$  is -2808 kJmol<sup>-1</sup>**

**Example** - Calculating the maximum non-expansion work of a reaction

**Method:** The non-expansion work available from the reaction is equal to the change in standard Gibbs energy for the reaction.

$$\Delta_r G = \Delta_r H - T\Delta_r S$$

The standard reaction Gibbs energy is then:

$$\begin{aligned}\Delta_r G &= -2808\text{kJmol}^{-1} - (310\text{K}) \times (18204\text{JK}^{-1}\text{mol}^{-1}) \\ &= -2865\text{kJmol}^{-1}\end{aligned}$$

## **Example** - Calculating the maximum non-expansion work of a reaction

$$w_{e, \max} = -2865 \text{ kJ}$$

for the combustion of 1 mol glucose molecules, and the reaction can be used to do up to 2865 kJ of non-expansion work.



## 3.8 Combining the first and second laws

### State functions

Temperature:  $T$

Pressure:  $p$

Volume:  $V$

The internal energy:  $U$

The entropy:  $S$

The enthalpy:  $H$

The Helmholtz energy:  $A$

The Gibbs energy:  $G$



## Five state functions in the First and Second Laws:

The internal energy:  $U$

The entropy:  $S$

The enthalpy:  $H$

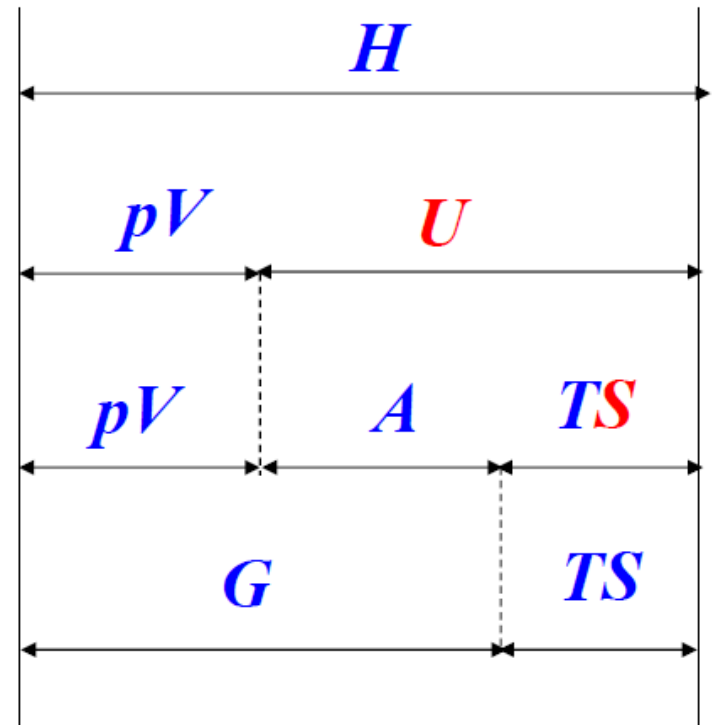
The Helmholtz energy:  $A$

The Gibbs energy:  $G$

$$H = U + pV$$

$$A = U - TS$$

$$G = H - TS = U + pV - TS = A + pV$$





## A. The fundamental equations

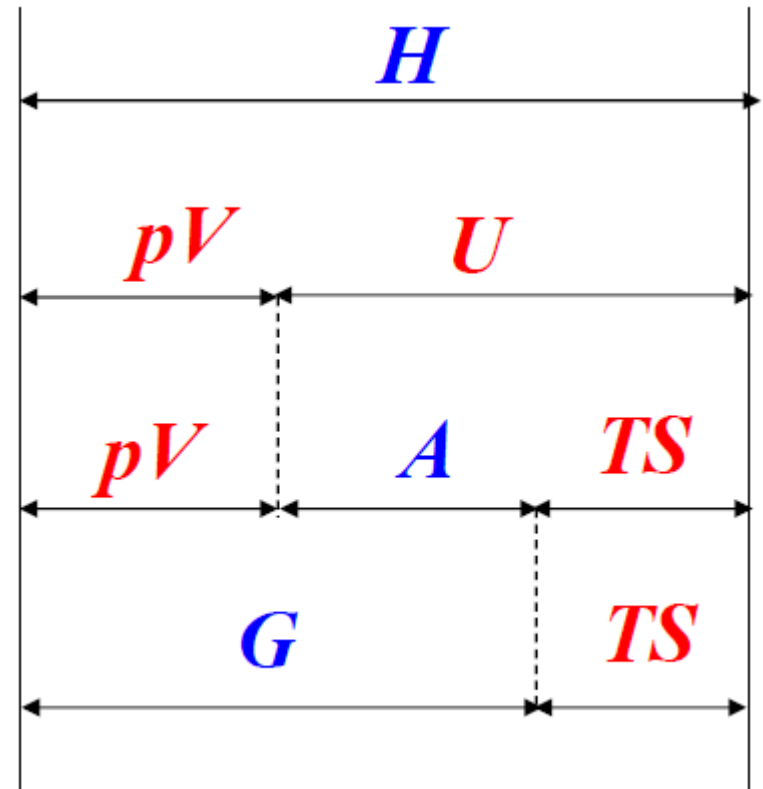
**The First Law:**  $dU = dq + dw$

**For a reversible change in a closed system of constant composition, and in the absence of any non-expansion work:**

$$dw = -pdV \text{ and } dq = TdS$$

$$dU = TdS - pdV$$

$dU$  is an exact differential, its value is independent of path.





$$dU = TdS - pdV$$

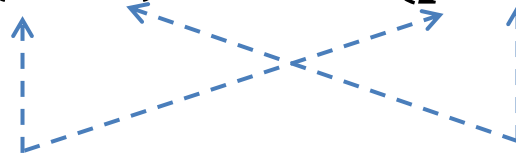


Thermal  
Equilibrium

Mechanical  
Equilibrium

$(T, S)$

$(p, V)$



Intensive properties

Extensive properties



$$H=U+pV, \quad A=U-TS, \quad G=U+pV-TS$$

$$dH = dU + pdV + Vdp$$

$$= (\cancel{TdS} - \cancel{pdV}) + \cancel{pdV} + Vdp$$

$$= TdS + Vdp$$

$$dG = dU + pdV + Vdp - TdS - SdT$$

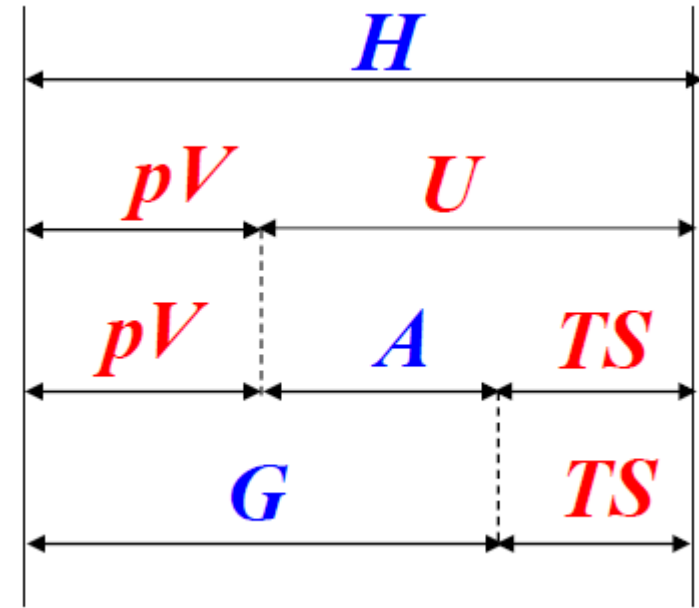
$$= (\cancel{TdS} - \cancel{pdV}) + \cancel{pdV} + Vdp - \cancel{TdS} - SdT$$

$$= Vdp - SdT$$

$$dA = dU - TdS - SdT$$

$$= (\cancel{TdS} - \cancel{pdV}) - \cancel{TdS} - SdT$$

$$= -p dV - SdT$$





$$dU = TdS - pdV$$

$$dH = TdS + Vdp$$

$$dG = Vdp - SdT$$

$$dA = -pdV - SdT$$

The fundamental equations

## 应用

对于U, H, S, A, G等热力学函数，只要其独立变量选择合适，就可以从一个已知的热力学函数通过偏微商，求得其他热力学函数，从而可以把一个均匀系统的平衡性质完全确定下来。

$$G = G(T, p)$$

$$dG = -SdT + Vdp$$

$$V = \left(\frac{\partial G}{\partial p}\right)_T$$

$$S = -\left(\frac{\partial G}{\partial T}\right)_p$$

$$H = G + TS = G - T\left(\frac{\partial G}{\partial T}\right)_p$$

$$U = H - pV = G - T\left(\frac{\partial G}{\partial T}\right)_p - p\left(\frac{\partial G}{\partial p}\right)_T$$

$$A = G - pV = G - p\left(\frac{\partial G}{\partial p}\right)_T$$



## B. The properties of the internal energy

### The Maxwell relations

$$\begin{aligned} z = f(x, y) \quad \quad \quad dz &= \left( \frac{\partial z}{\partial x} \right)_y dx + \left( \frac{\partial z}{\partial y} \right)_x dy \\ &= g dx + h dy \end{aligned}$$

$$\text{where } g = \left( \frac{\partial z}{\partial x} \right)_y, \quad h = \left( \frac{\partial z}{\partial y} \right)_x$$



The first derivative of  $g$  with respect to  $y$ , and  $h$  to  $x$

$$\mathbf{g} = \left( \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right)_y, \quad \mathbf{h} = \left( \frac{\partial \mathbf{z}}{\partial \mathbf{y}} \right)_x$$

$$\left. \begin{aligned} \left( \frac{\partial \mathbf{g}}{\partial \mathbf{y}} \right)_x &= \frac{\partial^2 \mathbf{z}}{\partial \mathbf{x} \partial \mathbf{y}} \\ \left( \frac{\partial \mathbf{h}}{\partial \mathbf{x}} \right)_y &= \frac{\partial^2 \mathbf{z}}{\partial \mathbf{y} \partial \mathbf{x}} \end{aligned} \right\} \left( \frac{\partial \mathbf{g}}{\partial \mathbf{y}} \right)_x = \left( \frac{\partial \mathbf{h}}{\partial \mathbf{x}} \right)_y$$



$$dU = TdS - pdV \quad \Rightarrow \quad U = U(S, V)$$

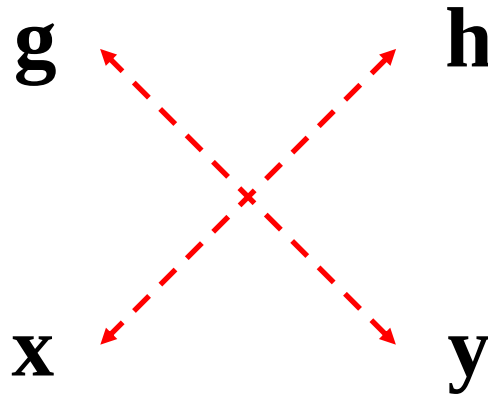
$$dU = \left( \frac{\partial U}{\partial S} \right)_V dS + \left( \frac{\partial U}{\partial V} \right)_S dV$$

$$g = \left( \frac{\partial U}{\partial S} \right)_V = T, \quad h = \left( \frac{\partial U}{\partial V} \right)_S = -p$$

$$\left( \frac{\partial g}{\partial y} \right)_x = \left( \frac{\partial h}{\partial x} \right)_y \quad \left( \frac{\partial T}{\partial V} \right)_S = - \left( \frac{\partial p}{\partial S} \right)_V$$



$$z = f(x, y) \quad dz = \left( \frac{\partial z}{\partial x} \right)_y dx + \left( \frac{\partial z}{\partial y} \right)_x dy$$
$$= gdx + hdy$$



$$\left( \frac{\partial g}{\partial y} \right)_x = \left( \frac{\partial h}{\partial x} \right)_y$$



$$dU = TdS - pdV \quad \left( \frac{\partial T}{\partial V} \right)_S = - \left( \frac{\partial p}{\partial S} \right)_V$$

$$dH = TdS + Vdp \quad \left( \frac{\partial T}{\partial p} \right)_S = \left( \frac{\partial V}{\partial S} \right)_p$$

$$dA = -pdV - SdT \quad \left( \frac{\partial p}{\partial T} \right)_V = \left( \frac{\partial S}{\partial V} \right)_T$$

$$dG = Vdp - SdT \quad \left( \frac{\partial V}{\partial T} \right)_p = - \left( \frac{\partial S}{\partial p} \right)_T$$

$$dz = gdx + hdy$$

$$g = (\partial z / \partial x)_y$$

$$h = (\partial z / \partial y)_x$$

$$\left( \frac{\partial g}{\partial y} \right)_x = \left( \frac{\partial h}{\partial x} \right)_y$$

**Maxwell  
relations**



$$dU = TdS - pdV$$

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$

$$\left(\frac{\partial U}{\partial S}\right)_V = T, \quad \left(\frac{\partial U}{\partial V}\right)_S = -p$$

$$dH = TdS + Vdp$$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

$$\left(\frac{\partial H}{\partial S}\right)_p = T, \quad \left(\frac{\partial H}{\partial p}\right)_S = V$$

$$dA = -p dV - SdT$$

$$\left(\frac{\partial p}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$$

$$\left(\frac{\partial A}{\partial V}\right)_T = -p, \quad \left(\frac{\partial A}{\partial T}\right)_V = -S$$

$$dG = Vdp - SdT$$

$$\left(\frac{\partial V}{\partial T}\right)_p = -\left(\frac{\partial S}{\partial p}\right)_T$$

$$\left(\frac{\partial G}{\partial p}\right)_T = V, \quad \left(\frac{\partial G}{\partial T}\right)_p = -S$$



## Maxwell方程的应用

- 可用容易由实验测定的偏微分来代替那些不易直接测定的偏微分
- 导出其它具有普遍意义的公式



## The variation of internal energy with volume

The internal pressure is defined as  $\pi_T = \left( \frac{\partial U}{\partial V} \right)_T$

Since  $dU = \left( \frac{\partial U}{\partial S} \right)_V dS + \left( \frac{\partial U}{\partial V} \right)_S dV$

If divide both sides of equation by  $dV$  with the constraint of constant  $T$

$$\left( \frac{\partial U}{\partial V} \right)_T = \left( \frac{\partial U}{\partial S} \right)_V \left( \frac{\partial S}{\partial V} \right)_T + \left( \frac{\partial U}{\partial V} \right)_S$$



$$\left(\frac{\partial U}{\partial V}\right)_T = \left(\frac{\partial U}{\partial S}\right)_V \left(\frac{\partial S}{\partial V}\right)_T + \left(\frac{\partial U}{\partial V}\right)_S \quad \leftarrow \quad \left(\frac{\partial U}{\partial S}\right)_V = T, \quad \left(\frac{\partial U}{\partial V}\right)_S = -p$$

$$= T \left(\frac{\partial S}{\partial V}\right)_T - p \quad \leftarrow \quad \left(\frac{\partial p}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$$

$$= T \left(\frac{\partial p}{\partial T}\right)_V - p$$

Therefore:

$$\underline{\underline{\pi_T = T \left(\frac{\partial p}{\partial T}\right)_V - p}}$$

## Example - Deriving a thermodynamic relation

Show thermodynamically that  $\pi_T = 0$  for a perfect gas, and compute its value for a van der Waals gas

**Method:** Proving a result 'thermodynamically' means basing it entirely on general thermodynamic relations and equations of state.

**Answer:** 1). For a perfect gas,  $pV = nRT$ , and

$$\left(\frac{\partial p}{\partial T}\right)_V = nR / V$$

From the equation of  $\pi_T = T \left( \frac{\partial p}{\partial T} \right)_V - p$

$$\pi_T = \frac{nRT}{V} - p = 0$$

## Example Deriving a thermodynamic relation

2). The equation of state of a van der Waals gas is

$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2} \quad \longrightarrow \quad \left( \frac{\partial p}{\partial T} \right)_V = \frac{nR}{V - nb}$$

That is

$$\begin{aligned} \pi_T &= T \left( \frac{\partial p}{\partial T} \right)_V - p \\ &= \frac{nRT}{V - nb} - \frac{nRT}{V - nb} + a \frac{n^2}{V^2} \\ &= a \frac{n^2}{V^2} \end{aligned}$$

## Comment

This result for  $\pi_T$  implies that the internal energy of a van der Waals gas increases when it expands isothermally (that is,  $(\partial U/\partial V)_T > 0$ ), and that the increase is related to the parameter  $a$ , which models the attractive interactions between the particles. A larger molar volume, corresponding to a greater average separation between molecules, implies weaker mean intermolecular attractions, so the total energy is greater.

## 3.9 Properties of the Gibbs energy

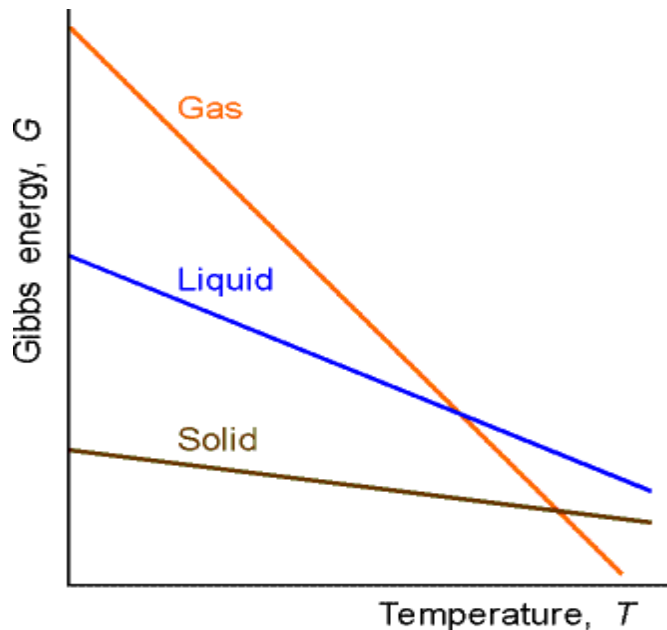
### A. General considerations

For a closed system doing non-expansion work and at constant composition

$$\begin{array}{ccc} d\mathbf{G} = Vd\mathbf{p} - SdT & \longrightarrow & \left( \frac{\partial V}{\partial T} \right)_p = - \left( \frac{\partial S}{\partial p} \right)_T \\ \updownarrow & & \\ \underline{G = G(p, T)} & & \left( \frac{\partial G}{\partial p} \right)_T = V, \quad \left( \frac{\partial G}{\partial T} \right)_p = -S \end{array}$$



$$G = G(p, T) \quad dG = Vdp - SdT \quad \left( \frac{\partial G}{\partial T} \right)_p = -S$$



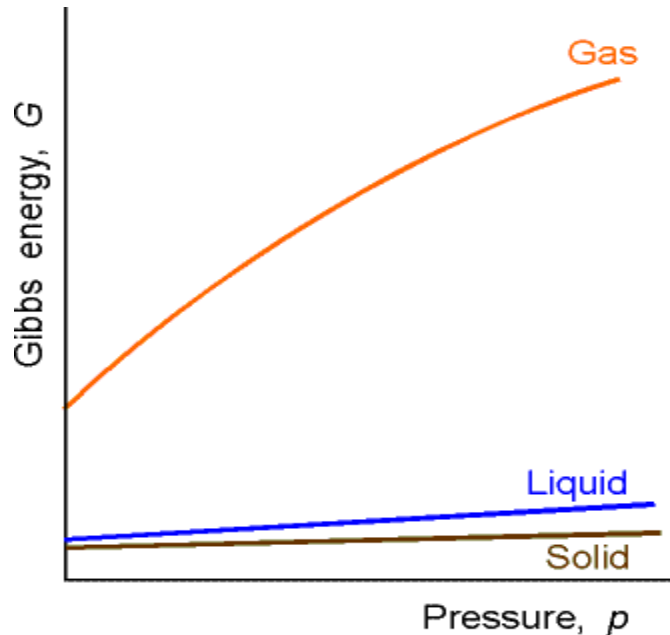
The variation of the Gibbs energy with the temperature is determined by the entropy

The entropy of the gaseous phase of a substance is greater than that of the liquid phase, and the entropy of the solid phase is smallest, the **Gibbs energy** changes most steeply for the **gas phase**, followed by the liquid phase, and then the solid phase of the substance.

$$S(g) > S(l) > S(s)$$



$$G = G(p, T) \quad dG = Vdp - SdT \quad \left( \frac{\partial G}{\partial p} \right)_T = V$$



The variation of the Gibbs energy with the pressure determined by the volume of the sample

The volume of the gaseous phase of a substance is greater than that of the same amount of liquid phase, and the volume of the solid phase is smallest, the Gibbs energy changes most steeply for the gas phase, followed by the liquid phase, and then the solid phase of the substance.

## **Example - Calculating the effect of pressure on the Gibbs energy**

**Calculate the change in the molar Gibbs energy of:**

- (a) liquid water treated as an incompressible fluid, and (b) water vapour treated as a perfect gas, when the pressure is increased isothermally from 1.0 bar to 2.0 bar at 298 K.**

## Example - Calculating the effect of pressure on the Gibbs energy

**Method:** In each case, the change in molar Gibbs energy can be obtained by integration of the equation  $dG = V dp - S dT$  with the temperature held constant:

$$G_m(p_f) - G_m(p_i) = \int_{p_i}^{p_f} V_m dp$$

## **Example-** Calculating the effect of pressure on the Gibbs energy

**Answer:**

**(a) For an incompressible fluid, the molar volume is independent of the pressure so  $V_m$  can be treated as a constant. For the incompressible liquid,  $V_m$  is constant at  $18.0 \text{ cm}^3\text{mol}^{-1}$ , then**

$$\begin{aligned} G_m(p_f) - G_m(p_i) &= V_m \int_{p_i}^{p_f} dp = V_m \times (p_f - p_i) \\ &= (18.0 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}) \times (1.0 \times 10^5 \text{ Pa}) \\ &= +1.8 \text{ Pa m}^3 \text{ mol}^{-1} = +1.8 \text{ J mol}^{-1} \end{aligned}$$

## Example - Calculating the effect of pressure on the Gibbs energy

Answer:

**(b)** For a perfect gas, the molar volume varies with pressure as  $V_m = RT/p$ , so this expression must be used in the integrand, and the integration performed treating  $RT$  as a constant:

$$\begin{aligned} G_m(p_f) - G_m(p_i) &= \int_{p_i}^{p_f} \frac{RT}{p} dp = RT \ln \left( \frac{p_f}{p_i} \right) \\ &= (2.48 \text{ kJ mol}^{-1}) \times \ln 2.0 \\ &= +1.7 \text{ kJ mol}^{-1} \end{aligned}$$

## Example - Calculating the effect of pressure on the Gibbs energy

**(a) For an incompressible**

$$G_m(p_f) - G_m(p_i) = +1.8 \text{ Jmol}^{-1}$$

**(b) For a perfect gas**

$$G_m(p_f) - G_m(p_i) = +1.7 \text{ kJmol}^{-1}$$



## B. The variation of the Gibbs energy with temperature

$$S = (H - G)/T \quad \left( \frac{\partial G}{\partial T} \right)_p = -S$$

$$\left( \frac{\partial G}{\partial T} \right)_p = \frac{G - H}{T} \quad \left( \frac{\partial G}{\partial T} \right)_p - \frac{G}{T} = -\frac{H}{T}$$

$$\left( \frac{\partial}{\partial T} \left( \frac{G}{T} \right) \right)_p = \frac{1}{T} \left( \frac{\partial G}{\partial T} \right)_p + G \frac{d}{dT} \frac{1}{T}$$

$$= \frac{1}{T} \left( \frac{\partial G}{\partial T} \right)_p - \frac{G}{T^2}$$

$$= \frac{1}{T} \left\{ \left( \frac{\partial G}{\partial T} \right)_p - \frac{G}{T} \right\}$$

$$\left( \frac{\partial}{\partial T} \left( \frac{G}{T} \right) \right)_p = -\frac{H}{T^2}$$

$$\left( \frac{\partial}{\partial T} \left( \frac{\Delta G}{T} \right) \right)_p = -\frac{\Delta H}{T^2}$$

the Gibbs-Helmholtz  
( $G-H$ ) equation



## the Gibbs-Helmholtz ( $G-H$ ) equation

$$\left( \frac{\partial}{\partial T} \left( \frac{\Delta G}{T} \right) \right)_p = -\frac{\Delta H}{T^2} \quad \Rightarrow \quad \frac{\Delta G_2}{T_2} - \frac{\Delta G_1}{T_1} = - \int_{T_1}^{T_2} \frac{\Delta H}{T^2} dT$$

It shows that if we know the enthalpy of the system, then we know how  $G/T$  varies with temperature.



## C. The variation of the Gibbs energy with pressure

At constant temperature:

$$(dG = Vdp - SdT)$$

$$G(p_f) = G(p_i) + \int_{p_i}^{p_f} Vdp$$

## Liquids and solids

For a liquid or solid, the volume changes only slightly as the pressure changes, so  $V$  may be treated as a constant.

For molar quantities:

$$\begin{aligned} G_{\text{m}}(p_{\text{f}}) &= G_{\text{m}}(p_{\text{i}}) + V_{\text{m}}(p_{\text{f}} - p_{\text{i}}) \\ &= G_{\text{m}}(p_{\text{i}}) + V_{\text{m}}\Delta p \end{aligned}$$

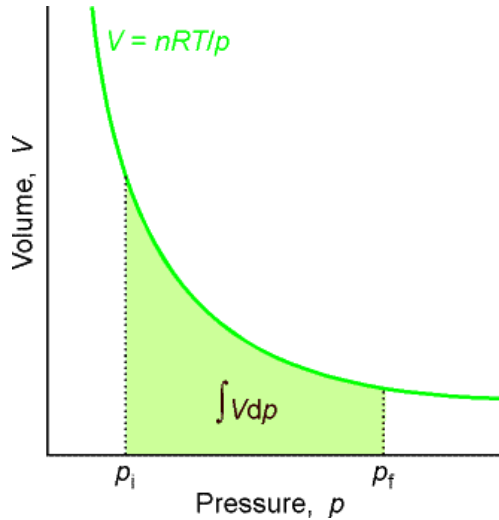
At normal lab. conditions  $V_{\text{m}}\Delta p$  is small, so it suggests that the  $G$  of solid and liquid are independent of  $p$ .

Hence, we may usually suppose that the Gibbs energies of solids and liquids are independent of pressure.

## Gases

For a perfect gas, the Gibbs energy may depend strongly on the pressure; and the volume also varies markedly with the pressure. Then  $V = nRT/p$ :

$$\begin{aligned} G(p_f) &= G(p_i) + \int_{p_i}^{p_f} V dp \\ &= G(p_i) + nRT \int_{p_i}^{p_f} \frac{dp}{p} \\ &= G(p_i) + nRT \ln \left( \frac{p_f}{p_i} \right) \end{aligned}$$



$$G(p_f) = G(p_i) + \int_{p_i}^{p_f} V dp$$

$$G(p_f) = G(p_i) + nRT \ln \left( \frac{p_f}{p_i} \right)$$

If set  $p_i = p^\ominus$ , (the standard pressure of 1 bar) then the Gibbs energy of a perfect gas at a pressure  $p$  is related to its standard value by

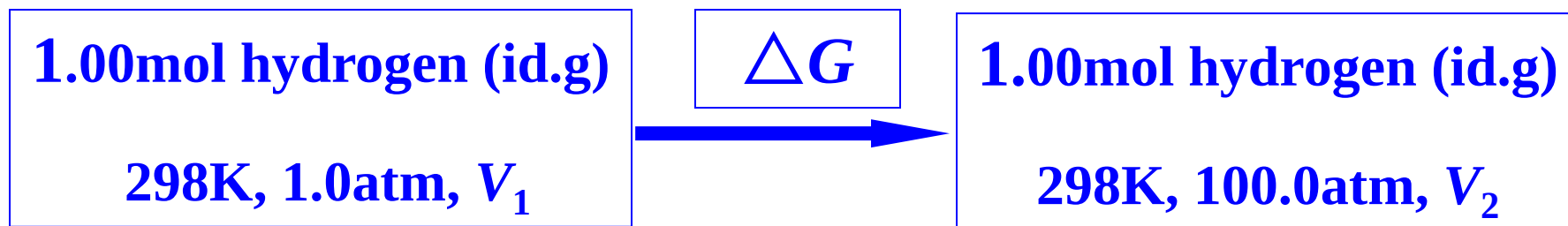
$$G(p) = G^\ominus + nRT \ln \left( \frac{p}{p^\ominus} \right)$$

## Example

**Calculate the change in the molar Gibbs energy of hydrogen when its pressure is increased isothermally from 1.0 atm to 100.0 atm at 298K.**

## Example

Answer:



$$G(p_f) = G(p_i) + nRT \ln \left( \frac{p_f}{p_i} \right)$$

$$\begin{aligned} \Delta G_m &= RT \ln \left( \frac{p_f}{p_i} \right) \\ &= (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K}) \times \ln \left( \frac{100.00 \text{ atm}}{1.00 \text{ atm}} \right) \\ &= 11.41 \text{ kJ mol}^{-1} \end{aligned}$$

## D. The standard Gibbs energies of reaction

The standard Gibbs energy of reaction can be based on the standard entropies and enthalpies of reaction:

$$\Delta_{\text{r}}G = \Delta_{\text{r}}H - T\Delta_{\text{r}}S$$

## The standard Gibbs energies of formation

The standard Gibbs energy of formation is the standard reaction Gibbs energy for the formation of a compound from its elements in their reference states.

$$\Delta_{\text{r}}G = \sum_{\text{Products}} \nu \Delta_{\text{f}}G - \sum_{\text{Reactants}} \nu \Delta_{\text{f}}G$$

$$\Delta_r G = \sum_{\text{Products}} \nu \Delta_f G - \sum_{\text{Reactants}} \nu \Delta_f G$$

Standard Gibbs energies of formation of the elements in their reference states are zero, because their formation is 'null' reaction.

**Illustration:** To calculate the standard Gibbs energy of the reaction  $\text{CO(g)} + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$  at 25°C:

$$\begin{aligned} \Delta_r G^\ominus &= \Delta_f G^\ominus(\text{CO}_2, \text{g}) - \left\{ \Delta_f G^\ominus(\text{CO}, \text{g}) + \frac{1}{2} \Delta_f G^\ominus(\text{O}_2, \text{g}) \right\} \\ &= -3394.4 - \left\{ -(137.2) + \frac{1}{2} (0) \right\} = -257.2 \text{ kJ mol}^{-1} \end{aligned}$$

### 3. 10 Real gases: the fugacity

At various stages in the development of physical chemistry it is necessary to switch from consideration of idealized systems to real systems. In many cases it is desirable to preserve the form of the expressions that have been derived for the idealized system. Then deviations from the idealized behavior can be expressed most simply. We shall illustrate such procedure in this section by considering how the expressions that have been derived for perfect gases, are adapted to describe real gases.



The fugacity coefficient,  $\varphi$ , is defined as:

$$\varphi = \frac{f}{p} \quad \text{or} \quad f = \varphi p$$

$p$ : the true pressure of real gas

$f$ : fugacity, the effective pressure

The fugacity coefficient,  $\varphi$  :

- 1). Dimensionless;
- 2). depends on the identity of the gas, the pressure, and the temperature.



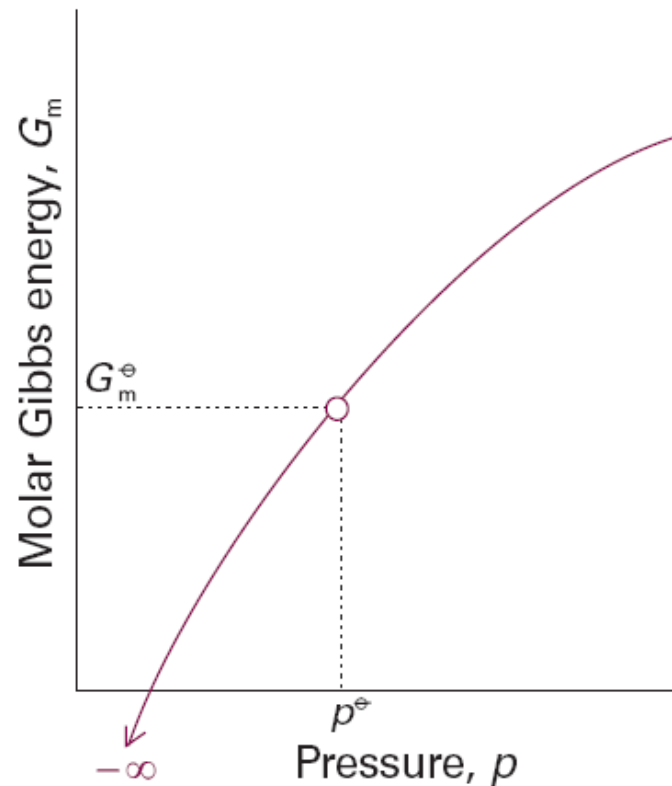
For a real gas, replace the true pressure,  $p$ , by an effective pressure,  $f$ , then the **fugacity** of a real gas is defined as:

$$G(p) = G^{\ominus} + nRT \ln \left( \frac{f}{p^{\ominus}} \right)$$

The name '**fugacity**' comes from the Latin for 'fleetness' in the sense of '**escaping tendency**'. Fugacity has the same dimension as pressure.



$$G(p) = G^{\ominus} + nRT \ln \left( \frac{p}{p^{\ominus}} \right)$$



$p \rightarrow 0$

The molar Gibbs energy becomes negatively infinite.



$$G(p_f) = G(p_i) + \int_{p_i}^{p_f} V dp$$

$$\int_{p'}^p V_m dp = G_m - G_m' = RT \ln \frac{f}{f'}$$

If the gas were perfect

$$\int_{p'}^p V_{perfect,m} dp = RT \ln \frac{p}{p'}$$

$$\int_{p'}^p (V_m - V_{perfect,m}) dp = RT \left( \ln \frac{f}{f'} - \ln \frac{p}{p'} \right)$$

$$\int_{p'}^p (V_m - V_{perfect,m}) dp = RT \ln \left( \frac{f}{f'} \times \frac{p'}{p} \right)$$



$$\int_{p'}^p (V_m - V_{perfect,m}) dp = RT \ln \left( \frac{f}{f'} \times \frac{p'}{p} \right)$$

$$\ln \left( \frac{f}{p} \times \frac{p'}{f'} \right) = \frac{1}{RT} \int_{p'}^p (V_m - V_{perfect,m}) dp$$

**When  $p' \rightarrow 0$ , the gas behaves perfectly and  $f'$  becomes equal to the pressure  $p'$ . Therefore,  $f'/p' \rightarrow 1$  as  $p' \rightarrow 0$ .**

$$\ln \frac{f}{p} = \frac{1}{RT} \int_0^p (V_m - V_{perfect,m}) dp$$

$$\ln \phi = \frac{1}{RT} \int_0^p (V_m - V_{perfect,m}) dp$$



$$\ln \phi = \frac{1}{RT} \int_0^p (V_m - V_{perfect,m}) dp$$

$$V_{perfect,m} = \frac{RT}{p} \quad V_m = \frac{RTZ}{p}$$

$$\ln \phi = \int_0^p \frac{Z - 1}{p} dp$$

where  $Z$  is the compression factor of the gas. This equation is an explicit expression for the **fugacity coefficient** at any pressure  $p$ . The fugacity of the gas at that pressure can be obtained by  $f = \phi p$ .



The perfect gases  $Z = \frac{pV_m}{RT} = 1$

---

Real gases

$Z > 1$   $V_m > V_m^\circ$  expansion

$Z < 1$   $V_m < V_m^\circ$  compression



$$\ln \varphi = \int_0^p (Z - 1) \frac{dp}{p}$$

$$\left(\frac{\partial G}{\partial p}\right)_T = V$$

If  $Z < 1$  throughout the range of integration:

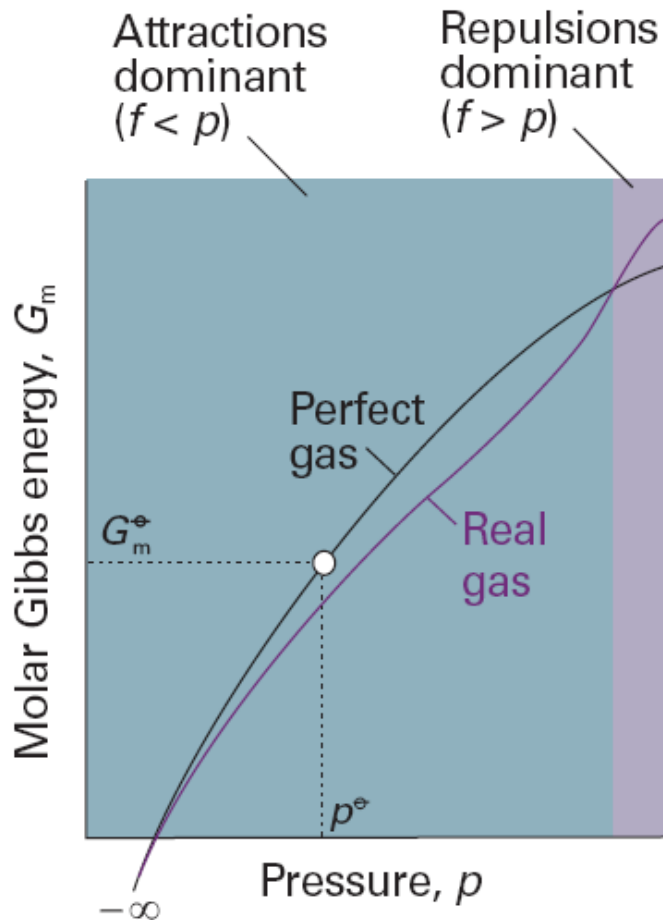
The integrand  $< 0$ ,  $\varphi < 1$ ,  $\Rightarrow f < p$ ,

The molecules tend to stick together and the molar Gibbs energy of the gas is less than that of a perfect gas;

If  $Z > 1$  throughout the range of integration (higher pressure):

The integrand  $> 0$ ,  $\varphi > 1$ ,  $\Rightarrow f > p$ ,

The repulsive interactions are dominant and tend to drive the particles apart, and the molar Gibbs energy of the gas is greater than that of a perfect gas



The  $G$  of a real gas

For a real gas:

$$G(p) = G^ominus + nRT \ln \left( \frac{p}{p^ominus} \right)$$

- at higher  $p$ ,  $G_{real} < G_{perfect}$   
the repulsive forces are dominant and the 'escaping tendency' is increased.
- at lower  $p$ ,  $G_{real} > G_{perfect}$   
the attractive forces are dominant and the molecules have a lower 'escaping tendency'.
- as  $p \rightarrow 0$ ,  $G$  coincides with the value for a perfect gas.

## Example - Calculating a fugacity

Suppose that the attractive interactions between gas particles can be neglected, and find an expression for the fugacity of a van der Waals gas in terms of the pressure. Estimate its value for ammonia at 10.00 atm and 298.15 K.

## Example - Calculating a fugacity

**Method:** The starting point for the calculation is equation

$$\ln \varphi = \int_0^p \left( \frac{Z - 1}{p} \right) dp$$

**Answer:** the van der Waals equation is as followings:

$$\begin{array}{l} p = \frac{RT}{V_m - b} - \frac{a}{V_m^2} \\ \downarrow \\ p = \frac{RT}{V_m - b} \end{array} \quad \longrightarrow \quad pV_m = RT + bp$$

and

$$Z = \frac{pV_m}{RT}$$

hence

$$Z = 1 + \frac{bp}{RT}$$

## Example - Calculating a fugacity

$$\begin{aligned}\ln \varphi &= \int_0^p \left( \frac{Z - 1}{p} \right) dp \\ &= \int_0^p \left( \frac{b}{RT} \right) dp = \frac{bp}{RT}\end{aligned}$$

$$\varphi = e^{bp/RT}$$

$$\underline{f = \varphi p = p e^{bp/RT}} \quad b = 3.71 \times 10^{-2} L \cdot mol^{-1}$$

For ammonia at 10.00 atm and 298.15 K

$$f = (10.00 \text{ atm}) \times e^{0.01515} = 10.2 \text{ atm}$$



# Summary

Reversible  $\longleftrightarrow$

Equilibrium

Irreversible  $\longleftrightarrow$

Spontaneous  
Non-spontaneous

q

Conditional



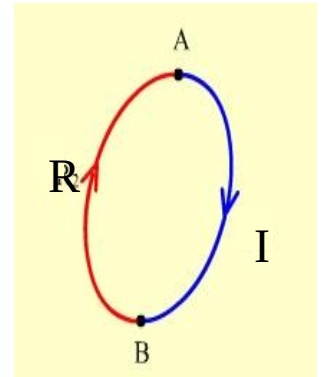
Unconditional

w



## S (entropy): State function $J \cdot K^{-1}$

$$dS = \frac{dq_{rev}}{T} \quad \Delta S = \int \frac{dq_{rev}}{T}$$



$$\text{Carnot: } \varepsilon = \frac{w}{q_h} = 1 + \frac{q_c}{q_h}; \varepsilon_{rev} = 1 - \frac{T_c}{T_h}$$

$\varepsilon \leq \varepsilon_{rev}$  : = reversible; < irreversible

$$\text{Clausius inequality: } \Delta S_{A \rightarrow B} \geq \left( \sum_A^B \frac{\delta q}{T} \right)$$

= reversible; > irreversible

$$\text{Adiabatic system: } q=0 \rightarrow \Delta S_{A \rightarrow B} \geq 0$$

= reversible; > irreversible

$$\text{Isolated system: } \Delta S_{A \rightarrow B} \geq 0$$

= reversible; > spontaneous



## Phase equilibrium

$$\Delta_{trs}S = \frac{\Delta_{trs}H}{T_{trs}}$$

## Isothermal (reversible) expansion

$$\Delta S = nR \ln \frac{V_f}{V_i} = nR \ln \frac{p_i}{p_f}$$

## Change T at constant p

$$S(T_f) = S(T_i) + C_p \ln \frac{T_f}{T_i}$$

## Change T at constant V

$$S(T_f) = S(T_i) + C_V \ln \frac{T_f}{T_i}$$



- Third law of thermodynamics

$$S(0\text{ K}) = 0$$

The Third-Law entropy, standard entropy  $S^\ominus(T)$

The standard reaction entropy  $\Delta_r S^\ominus$

$$\Delta_r S^\ominus = \sum_{\text{Products}} \nu S^\ominus - \sum_{\text{Reactants}} \nu S^\ominus$$



- **Helmholtz function**

*At constant temperature and volume*

$$A = U - TS$$

$$\Delta A \leq w \quad w: \text{non-expansion work}$$

$$w_{max} = \Delta A \quad \text{work function}$$

No non-expansion work

$$dA_{T,V} \leq 0 \quad \left\{ \begin{array}{l} < 0 \text{ spontaneous} \\ = 0 \text{ equilibriums} \end{array} \right.$$

Surrounding entropy change    System entropy change

$$dA = \boxed{dU} - \boxed{TdS} \leq 0$$

- **Gibbs Energy**

*At constant temperature and pressure*

$$G = H - TS$$

$$\Delta G \leq w \quad w: \text{non-expansion work}$$

$$w_{max} = \Delta G \quad \text{work function}$$

No non-expansion work

$$dG_{T,p} \leq 0 \quad \left\{ \begin{array}{l} < 0 \text{ spontaneous} \\ = 0 \text{ equilibriums} \end{array} \right.$$

Surrounding entropy change    System entropy change

$$dG = \boxed{dH} - \boxed{TdS} \leq 0$$

The enthalpy of a perfect gas is related to its internal energy by using  $pV = nRT$  in the definition of  $H$ :

$$**$H = U + pV = U + nRT$**$$

This relation implies that the change of enthalpy in reaction that produces or consumes gas is:

$$**$\Delta_r H = \Delta_r U + \Delta n_g RT$**$$



- Five state functions in the First and Second Laws

The internal energy:  $U$

The entropy:  $S$

The enthalpy:  $H$

The Helmholtz energy:  $A$

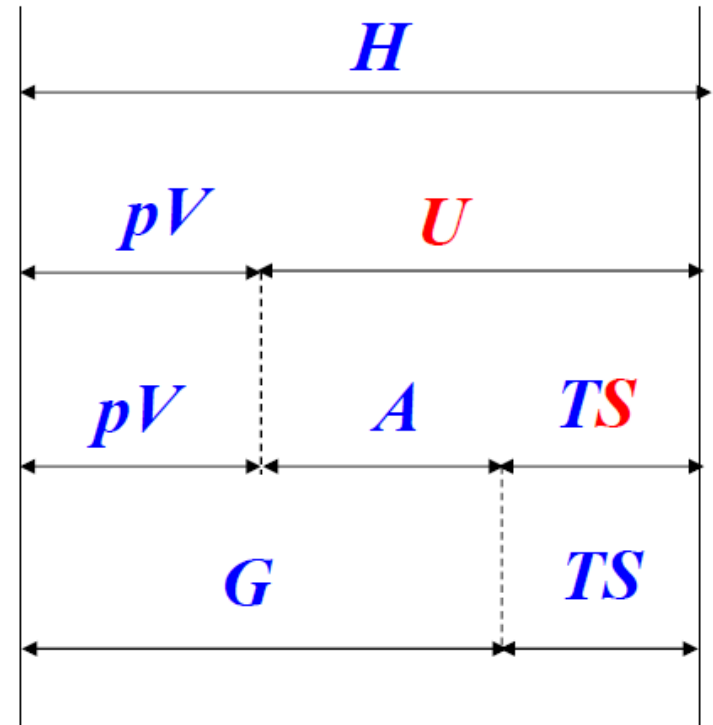
The Gibbs energy:  $G$

$$H = U + PV$$

$$G = A + PV$$

$$A = U - TS$$

$$G = H - TS$$





$$dU = TdS - pdV$$

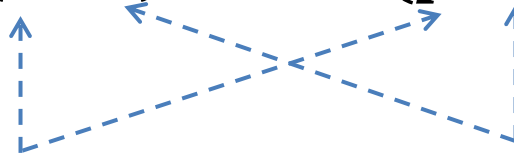


Thermal  
Equilibrium

Mechanical  
Equilibrium

$(T, S)$

$(p, V)$



Intensive properties

Extensive properties



$$dU = TdS - pdV$$

$$dH = TdS + Vdp$$

$$dG = Vdp - SdT$$

$$dA = -pdV - SdT$$



- The Maxwell relations

$$dz = gdx + hdy$$

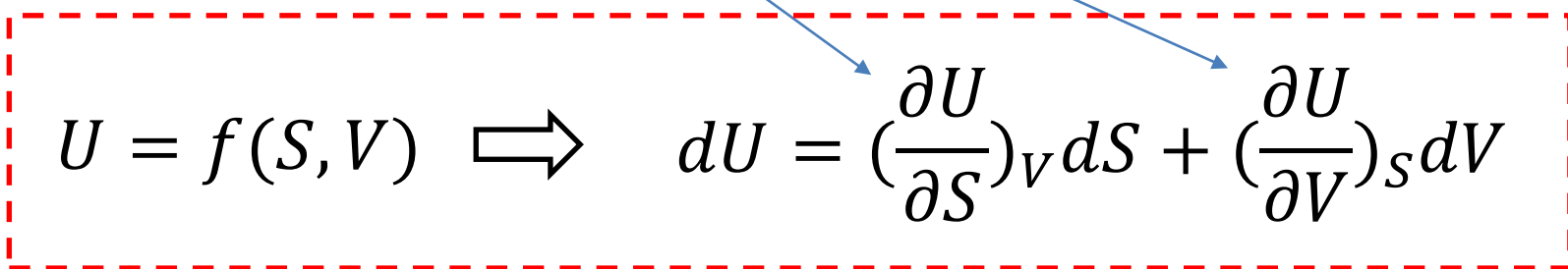
$$g = (\partial z / \partial x)_y$$

$$h = (\partial z / \partial y)_x$$

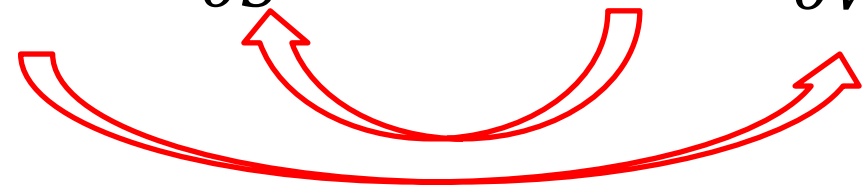
$$\left( \frac{\partial g}{\partial y} \right)_x = \left( \frac{\partial h}{\partial x} \right)_y$$



$$dU = TdS - pdV$$


$$U = f(S, V) \implies dU = \left(\frac{\partial U}{\partial S}\right)_V dS + \left(\frac{\partial U}{\partial V}\right)_S dV$$

$$T = \left(\frac{\partial U}{\partial S}\right)_V \quad -p = \left(\frac{\partial U}{\partial V}\right)_S$$


$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$



$$dU = TdS - pdV$$

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$

$$\left(\frac{\partial U}{\partial S}\right)_V = T, \quad \left(\frac{\partial U}{\partial V}\right)_S = -p$$

$$dH = TdS + Vdp$$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

$$\left(\frac{\partial H}{\partial S}\right)_p = T, \quad \left(\frac{\partial H}{\partial p}\right)_S = V$$

$$dA = -pdV - SdT$$

$$\left(\frac{\partial p}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$$

$$\left(\frac{\partial A}{\partial V}\right)_T = -p, \quad \left(\frac{\partial A}{\partial T}\right)_V = -S$$

$$dG = Vdp - SdT$$

$$\left(\frac{\partial V}{\partial T}\right)_p = -\left(\frac{\partial S}{\partial p}\right)_T$$

$$\left(\frac{\partial G}{\partial p}\right)_T = V, \quad \left(\frac{\partial G}{\partial T}\right)_p = -S$$



- Properties of the Gibbs energy

$$G = f(T, p)$$

$$dG = Vdp - SdT$$

$$\left(\frac{\partial G}{\partial p}\right)_T = V \qquad \left(\frac{\partial G}{\partial T}\right)_p = -S$$



- The Gibbs-Helmholtz ( $G$ - $H$ ) equation

$$\left( \frac{\partial}{\partial T} \left( \frac{\Delta G}{T} \right) \right)_p = -\frac{\Delta H}{T^2} \quad \Rightarrow \quad \frac{\Delta G_2}{T_2} - \frac{\Delta G_1}{T_1} = - \int_{T_1}^{T_2} \frac{\Delta H}{T^2} dT$$

$$H = \left( \frac{\partial (G/T)}{\partial (1/T)} \right)_p$$

It shows that if we know the enthalpy of the system, then we know how  $G/T$  varies with temperature.



- The variation of the Gibbs energy with pressure

Liquids and solids

$$\Delta G_m = V_m \Delta p$$

Gases

$$\Delta G_m = RT \ln \frac{p_f}{p_i}$$

$$G(p) = G^\ominus + nRT \ln \left( \frac{p}{p^\ominus} \right)$$



- The standard Gibbs energies of reaction**

Standard Gibbs energies of formation  $\Delta_f G^\ominus$

The standard Gibbs energy of reaction  $\Delta_r G^\ominus$

$$\Delta_r G^\ominus = \sum_{\text{Products}} v \Delta_f G^\ominus - \sum_{\text{Reactants}} v \Delta_f G^\ominus$$



- Real gas and fugacity

$$\varphi = \frac{f}{p}$$

$\varphi$ : fugacity coefficient  
 $f$ : fugacity

$$G(p) = G^\ominus + nRT \ln \left( \frac{f}{p^\ominus} \right)$$

$$\ln \varphi = \int_0^p (Z - 1) \frac{dp}{p}$$

$Z < 1$ , attractive interactions,  $G_{\text{real}} < G_{\text{perfect}}$

$Z > 1$ , repulsion interactions,  $G_{\text{real}} > G_{\text{perfect}}$



1. 概念： 可逆过程、自发过程、 $\Delta S$   $\Delta A$   $\Delta G$ 在特定条件下与功和热的关系
2. 三个判据

$$\Delta S \geq 0 \text{ isolated system}$$

$$\Delta A \leq 0 \text{ constant } T, V; w = 0$$

$$\Delta G \leq 0 \text{ constant } T, p; w = 0$$

3. 计算 $\Delta S, \Delta G$